

Spend Bits Where Queries Look: KV Cache Vector Quantization with Attention-Preserving Transforms

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Long-context LLM decoding reads the key-value (KV) cache at every step. Loading it takes longer than computing attention over it, so throughput is bandwidth-bound. Hence, reducing the cache size can raise both decoding speed and serving capacity. The challenge is to reduce cache size while preserving the attention products, keeping reconstruction cheap, and using a fixed per-token bit count. At two bits per element, the most competitive methods rely on orthogonal transforms. However, existing techniques are either data-oblivious or use the query statistics without deriving the transform from a distortion criterion. Moreover, they rely on transforms built on top of random or Hadamard rotations, which equalize variances across entries rather than compacting energy, and fixed-width scalar quantizers, which are suboptimal at low rates. In this paper, we formulate KV cache quantization as a transform coding problem in which distortion is the error in the attention products. We derive closed-form optimal transforms for keys and values from calibration statistics, under a high-resolution model. We show that the optimal key transform is not orthogonal and satisfies a generalized Parseval relation: the attention-aware distortion becomes mean-squared error (MSE) in the transform domain. Thus, we can use MSE-optimal vector quantizers applied directly to the transformed key coefficients. To meet the fixed-width layout requirement, we show that grouping coefficients into equal-volume partitions makes equal-size codebooks attain the variable-rate optimum under the same high-resolution model. At two bits per element, our method, termed NOVA-KV, recovers most of the long-context retrieval accuracy lost by scalar quantization methods at comparable throughput. The margin is widest on hybrid-attention mixture-of-experts models, an increasingly common design: on GPT-OSS-20B, prior two-bit transforms collapse at every context length, while NOVA-KV remains effective.

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Code: <https://github.com/Amir-zsh/nova-kv>

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1 Introduction

The key-value (KV) cache of a large language model (LLM) stores, for each past token and attention head, a key vector and a value vector, to avoid recomputation during decoding (Pope et al., 2023). The size of the cache is a central bottleneck in inference (Shazeer, 2019): it grows with context length and batch size, yet every decoding step reads it from memory in full (Sadhukhan et al., 2025) (Fig. 1). Since kernels compute attention products faster than the KV cache can be loaded from memory (Dao et al., 2022), decoding throughput is bound by memory bandwidth rather than by compute (Kwon et al., 2023). Thus, *KV cache compression* can reduce both per-step memory traffic and footprint, raising decoding speed and serving capacity.

Three requirements constrain the design of KV cache compressors. First, task accuracy must be preserved. Since capturing the effect of quantization on task accuracy is not straightforward, we can model the problem using classical rate-distortion (RD) theory (Berger, 1971), with the mean squared error (MSE) of the quantized values as the distortion metric. However, the cache contents are used to compute attention products, so we should minimize errors in the products rather than MSE in the factors (Zandieh et al., 2025; Zhou et al.,

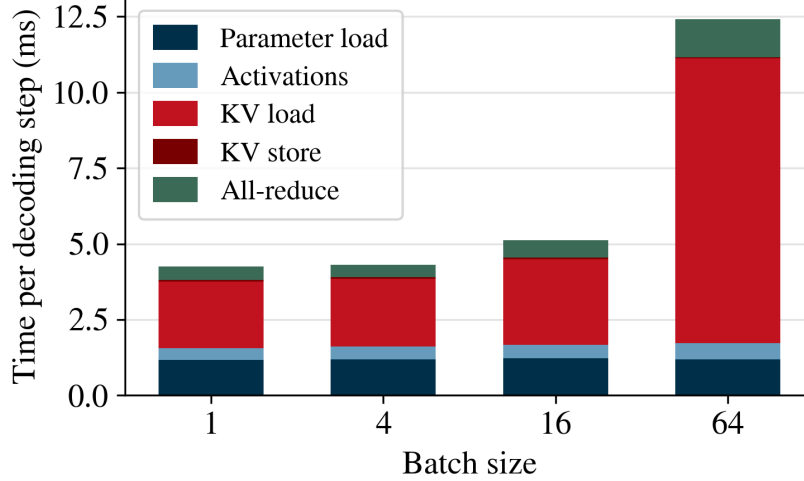


Figure 1 Measured per-step decode time for Qwen3-8B ($8 \times H100$, BF16, $S=16384$). At large batch sizes, KV cache loading dominates over parameter loading and compute.

2026). Second, complexity on the read path (Williams et al., 2009) has to be kept low, because the cache is read at every decoding step, so any per-element cost is incurred across the whole cache at every step. Third, a fixed-width layout must be used, so that tokens occupy a fixed number of bits and can be efficiently retrieved by serving engines (Kwon et al., 2023).

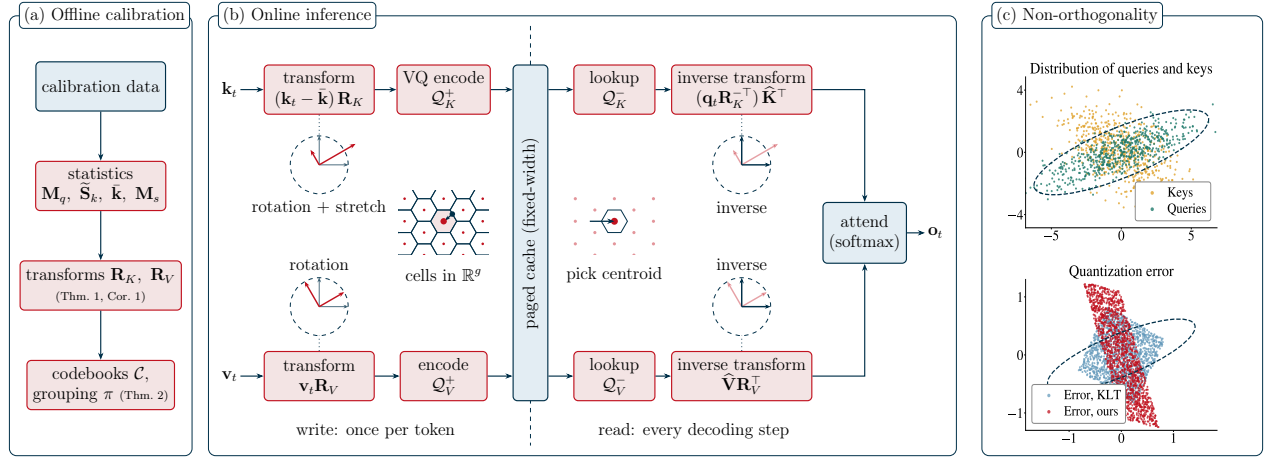


Figure 2 NOVA-KV. (a) Calibration yields the transforms, the volume-equalized grouping π , and the codebooks. (b) Online, each new key and value is transformed and encoded once on the write path; the read path, paid at every decoding step, is a table lookup. The dashed line separates the two paths. (c) Top: keys, queries, and query covariance (dashed). Bottom: quantization error at equal rate; our transform steers the error away from the high-energy query directions, preserving the attention logits.

Fixed-width quantization meets the last two requirements: decoding is a scaling or table lookup (Gray and Neuhoﬀ, 2002), and each token occupies a fixed number of bits. As the context grows, the KV cache dominates the memory footprint, so the reduction in inference time approaches the KV compression ratio itself. However, to preserve task accuracy, direct quantization of KV entries requires at least 4 bits per element (Liu et al., 2024b; Sheng et al., 2023); at 2 bits, accuracy degrades (Zhou et al., 2026). Transform coding (Goyal, 2001) has been proposed as an alternative to address this problem (Zandieh et al., 2025; Zhou et al., 2026).

In classical transform coding, a *data-dependent* linear operator (the transform) decorrelates the inputs; scalar quantization is applied to the outputs, followed by entropy coding. Because these transforms achieve

energy compaction, bits can be allocated proportionally to the coefficient variances, achieving lower overall distortion for a given average rate compared to directly encoding without a transform (Goyal, 2001). Existing transform-based KV cache methods depart from this classical scheme. In some cases, they use transforms, such as random or Hadamard rotations (Ashkboos et al., 2024; Zandieh et al., 2025), that are not optimized for the data. While data-driven transforms have been proposed, these are assembled heuristically from second-order statistics rather than as solutions to a specific rate-distortion optimization problem (Zhou et al., 2026). Finally, existing designs use fixed-rate scalar quantization, with transforms that *flatten* the variances so that the same rate can be applied for every vector entry; flattening removes the variance spread (energy compaction) that bit allocation exploits, so the fixed-width layout can be achieved but at the expense of reduced coding efficiency. Scalar quantization with variable-length entropy coding is effective (even at low rates) (Goyal, 2001), but requires sequential decoding, breaking the fixed-width layout and the parallel read path requirements.

In this paper, we formulate KV cache compression as a transform coding problem (Goyal, 2001) where distortion is set to be the attention product error. This error separates into two terms, one for the keys and one for the values, recovering the two criteria that prior work adopts separately (Zandieh et al., 2025; Zhou et al., 2026). We derive, in closed form, the optimal transform for each distortion term in the high-resolution regime. For *values*, the optimal transform is orthogonal and is obtained from the same covariance that prior work introduced as a heuristic (Zhou et al., 2026). For *keys*, our focus in this work, we show that the optimal transform is not orthogonal. All prior designs used orthogonal transforms, which we show are suboptimal in practice when the goal is to minimize the key-query inner product.

The transform we derive for the keys satisfies a generalized Parseval relation: the MSE in the transform domain equals the key-query inner-product distortion in the original domain. This relation reduces the design of the quantizer to a standard MSE problem, so any MSE-optimal quantizer can act on the transform coefficients. Based on this reduction, we propose vector quantization of the key coefficients with an optimized grouping of entries into vectors. Transform coding would normally allocate more bits to high-variance entries, resulting in codebooks of different sizes, which breaks the fixed-width layout that serving engines require. Instead, our grouping approximately equalizes group volumes, allowing equal-size codebooks to approach the variable-rate distortion bound (cf. Thm. 2). This combines the advantage of flattening (fixed bits per quantized input), while benefiting from energy compaction (fewer bits per input to achieve same distortion).

We integrate our method, termed NOVA-KV (Non-Orthogonal Vector-quantized Attention for KV cache), into a production serving stack (Zheng et al., 2024) with a fused decoding kernel (Tillet et al., 2019) (Fig. 2). At 2 bits per element, NOVA-KV recovers most of the accuracy lost by scalar methods in long-context retrieval. In terms of speed, decoding is on par with the scalar baseline (Zhou et al., 2026) at long contexts.

Our contributions are: 1) data-driven transforms that minimize a high-resolution bound on the attention product distortion (Sec. 3), 2) an analysis of vector quantization for the keys in the KV cache, including the optimal fixed-rate grouping of transform coefficients under an independent-Gaussian high-resolution model (Sec. 4), and 3) a serving-compatible implementation with a fused decoding kernel, which we test against state-of-the-art 2-bit methods (Sec. 5).

2 Preliminaries

Notation. Uppercase and lowercase bold letters, such as $\mathbf{A}$ and $\mathbf{a}$, denote matrices and vectors, respectively. The n th entry of $\mathbf{a}$ is a_n , and the (i, j) th entry of $\mathbf{A}$ is A_{ij} . Regular letters denote scalar values. We use row-vector notation and assume the eigenvalues are sorted in decreasing order. We use $(\cdot)^+$ to denote the pseudoinverse.

2.1 Attention and the KV Cache

Consider a single attention head (Vaswani et al., 2017) acting on a sequence of hidden states $\{\mathbf{x}_t\}_{t=1}^T$, $\mathbf{x}_t \in \mathbb{R}^{1 \times d}$, with projection weights $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d \times d}$, where d denotes the head dimension. The *query*, *key*, and *value* vectors are $\tilde{\mathbf{q}}_t = \mathbf{x}_t \mathbf{W}_Q$, $\tilde{\mathbf{k}}_t = \mathbf{x}_t \mathbf{W}_K$, $\mathbf{v}_t = \mathbf{x}_t \mathbf{W}_V$. We use rotary position embeddings (RoPE) (Su et al., 2024): queries and keys are rotated by a position-dependent block-diagonal matrix $\Phi_t \in \mathbb{R}^{d \times d}$ before the inner product, $\mathbf{q}_t = \tilde{\mathbf{q}}_t \Phi_t$ and $\mathbf{k}_t = \tilde{\mathbf{k}}_t \Phi_t$, while values carry no positional encoding. Symbols without a tilde denote

post-RoPE vectors. These are stacked into $\mathbf{Q} = [\mathbf{q}_1; \dots; \mathbf{q}_T]$, $\mathbf{K} = [\mathbf{k}_1; \dots; \mathbf{k}_T]$, $\mathbf{V} = [\mathbf{v}_1; \dots; \mathbf{v}_T] \in \mathbb{R}^{T \times d}$. The scores and outputs are:

$$\mathbf{S} = \text{softmax}_{\text{row}}(\mathbf{Q}\mathbf{K}^\top / \sqrt{d}) \in \mathbb{R}^{T \times T}, \quad \mathbf{O} = \mathbf{S}\mathbf{V}. \quad (1)$$

During autoregressive decoding, generating token $t + 1$ requires the query $\mathbf{q}_{t+1}$ to attend over all previous positions. To avoid recomputing past projections, the *KV cache* stores in memory $\mathbf{K}_{1:t} = [\mathbf{k}_1; \dots; \mathbf{k}_t]$ and $\mathbf{V}_{1:t} = [\mathbf{v}_1; \dots; \mathbf{v}_t]$.

2.2 Transform coding

Classical transform coding (Goyal, 2001) consists of three stages: a transform that produces uncorrelated coefficients, scalar quantization of each coefficient, and entropy coding of the indices. Entropy decoding is sequential and cannot be easily parallelized, so it is not well-suited for our problem, where the cache is read at every decoding step, and parallel decoding is needed to maintain high throughput.

Transform. By Parseval’s relation, an orthogonal $\mathbf{R}$ preserves the MSE, $\|\mathbf{x} - \hat{\mathbf{x}}\|_2 = \|\mathbf{x}\mathbf{R} - \mathcal{Q}(\mathbf{x}\mathbf{R})\|_2$, so the quantizer can be designed in the transform domain; for Gaussian sources in the high-resolution regime the Karhunen-Loève transform (KLT, the eigenbasis of the covariance), is optimal (Goyal, 2001; Gersho and Gray, 1992). A transform achieves *energy compaction* when only a few coefficients in the transform have high variance. The KLT is optimal in the sense that its leading p coefficients capture the most variance. Thus, we can find the KLT, $\mathbf{R}^*$, by optimizing the rank- p reconstruction error: given centered $\mathbf{x}_j$, $j = 1, \dots, M$, let

$$\mathbf{R}_p^* = \arg \min_{\mathbf{R} \in \mathbb{R}^{d \times p}} \sum_{j=1}^M \|\mathbf{x}_j - \mathbf{x}_j \mathbf{R} \mathbf{R}^+\|_2^2. \quad (2)$$

The KLT can be computed via eigendecomposition of the covariance matrix, and $\mathbf{R}_p^*$ is the KLT truncated to the p eigenvectors corresponding to the dominant eigenvalues of the data covariance. We will show in Sec. 3.1 that this is not optimal in our setting.

Quantization. A vector quantizer operates on vectors of dimension g and with an average rate b bits per entry, spending gb bits per vector. An encoder $\mathcal{Q}^+ : \mathbb{R}^{1 \times g} \rightarrow \{1, \dots, 2^{gb}\}$ assigns each subvector the index of its nearest codeword, and a decoder $\mathcal{Q}^-$ returns that codeword from a codebook $\mathcal{C} \subset \mathbb{R}^{1 \times g}$ of 2^{gb} entries; their composition is $\mathcal{Q}$ (Lloyd, 1982; Gersho and Gray, 1992). *Scalar quantization* (SQ), $g = 1$, acts on each coordinate independently. For a given rate, *vector quantization* (VQ), $g > 1$, achieves distortion no larger than SQ, since the products of scalar codebooks are valid vector codebooks (Lookabaugh and Gray, 1989). Classical transform coding pairs SQ with variable-length entropy coding; however, as discussed earlier, entropy coding prevents parallel decoding, reducing the efficiency of the read path and impacting throughput.

As a summary, given an invertible $\mathbf{R} \in \mathbb{R}^{d \times d}$ and a vector $\mathbf{x}$, transform coding encodes it as $\mathcal{Q}^+(\mathbf{x}\mathbf{R})$ and reconstructs it as $\hat{\mathbf{x}} = \mathcal{Q}(\mathbf{x}\mathbf{R}) \mathbf{R}^{-1}$.

2.3 Related work

KV cache reduction can be achieved by combining several complementary approaches: (i) token eviction, where cached tokens are dropped based on their estimated importance to future attention (Zhang et al., 2023; Li et al., 2024; Xiao et al., 2024; Devoto et al., 2025); (ii) low-rank designs, which shrink the head dimension (Liu et al., 2024a); and (iii) quantization (our approach).

Scalar KV quantization methods quantize entries independently and do not consider the attention products. Some designs perform per-channel quantization of keys and per-token quantization of values without accounting for input data statistics (Liu et al., 2024b). Others fit the quantizers to the data, typically learning per-channel codebooks from the cache statistics (Hooper et al., 2024; Cai et al., 2025), or boosting sensitive channels at the cost of per-channel metadata and non-uniform layouts (Xia et al., 2025).

Transform coding. QuaRot (Ashkboos et al., 2024) uses random or Hadamard rotations, and TurboQuant (Zandieh et al., 2025) debiases inner products. SpinQuant (Liu et al., 2025) and RotateKV (Su et al., 2025)

Table 1 KV cache compressors. *Transform*: linear map applied before quantization. *Criterion*: the objective the transform is derived from. *Target*: the resource reduced; bandwidth-oriented methods (bw) keep the read path to a scaling or a lookup, whereas storage-oriented methods (sto) pay a penalty in the read path. Had. stands for Hadamard, orth. stands for orthogonal.

Method	Transform	Criterion	Quant.	Target
KIVI	None	Heuristic	SQ	sto
KVQuant	None	Heuristic	SQ	sto
Kitty	None	Heuristic	SQ	sto
CommVQ	None	key MSE	VQ	sto
QuaRot	rand. Had. orth.	Heuristic	SQ	bw
TurboQuant	rand. orth.	inner-prod	SQ	bw
RotateKV	Had.+calib. orth.	Heuristic	SQ	bw
OSCAR	calib.+Had. orth.	Heuristic	SQ	bw
NOVA-KV	calib. non-orth.	attn. products	VQ	bw

learn the rotation on calibration data. Zhou et al. (2026) reweight the key error by query statistics. All are orthogonal and paired with SQ. We derive the transform from the RD problem using the attention-product distortion and show that the optimal key transform is non-orthogonal.

Vector quantization. Unlike model weights (Chee et al., 2023), KV caches are generated online and read at every decoding step, which changes the VQ requirements. Zhang et al. (2024) explores joint codebooks across channels. CommVQ (Li et al., 2025) targets memory *storage* using VQ on the keys with large codebooks whose reconstruction requires a matrix product paid over the whole cache at every step. We target bandwidth-bound serving by applying VQ to small groups of transform-domain entries and restricting reconstruction to fixed-width codebook lookups that can be fused into the attention kernel. Thus, our design jointly provides an attention-derived transform, fixed-width VQ, and a lookup-based reconstruction. In summary, NOVA-KV is the only method that targets attention products and optimizes both the transform and quantization for this purpose (see Table 1).

3 Attention-preserving transforms

We aim to design transforms $(\mathbf{R}_K, \mathbf{R}_V)$ and quantizers $(\mathcal{Q}_K(\cdot), \mathcal{Q}_V(\cdot))$ to compress keys and values

$$\hat{\mathbf{K}} = \mathcal{Q}_K(\mathbf{K}\mathbf{R}_K) \mathbf{R}_K^{-1}, \quad \hat{\mathbf{V}} = \mathcal{Q}_V(\mathbf{V}\mathbf{R}_V) \mathbf{R}_V^{-1}, \quad (3)$$

while minimizing the attention output error $\|\mathbf{S}\mathbf{V} - \hat{\mathbf{S}}\hat{\mathbf{V}}\|_F^2$, where $\hat{\mathbf{S}}$ is obtained by replacing $\mathbf{K}$ by $\hat{\mathbf{K}}$ in (1). In the high-resolution regime, second order terms on the quantization errors vanish, and the target error can be upper-bounded by the sum of two terms (Appendix E.3):

$$\|\mathbf{S}\mathbf{V} - \hat{\mathbf{S}}\hat{\mathbf{V}}\|_F \lesssim \|\mathbf{S} - \hat{\mathbf{S}}\|_F \|\mathbf{V}\|_2 + \|\mathbf{S}(\mathbf{V} - \hat{\mathbf{V}})\|_F. \quad (4)$$

We use the score error $\|\mathbf{S} - \hat{\mathbf{S}}\|_F$ as a criterion to derive $\mathbf{R}_K$ and the value error weighted by the scores $\|\mathbf{S}(\mathbf{V} - \hat{\mathbf{V}})\|_F$ as a criterion to derive $\mathbf{R}_V$. $\mathcal{Q}_K$ consists of VQs with fixed codebook sizes applied to subvectors (Sec. 4), while $\mathcal{Q}_V$ applies SQ to each transformed entry (Zhou et al., 2026).

3.1 Key transform

Since row-wise softmax is $1/2$ -Lipschitz (Gao and Pavel, 2017), the key-dependent term in (4), $\|\mathbf{S} - \hat{\mathbf{S}}\|_F$, can be replaced by the logit error: $\|\mathbf{S} - \hat{\mathbf{S}}\|_F \leq 1/(2\sqrt{d})\|\mathbf{Q}\mathbf{K}^\top - \mathbf{Q}\hat{\mathbf{K}}^\top\|_F$. Then, defining $\mathbf{M}_q = \mathbf{Q}^\top \mathbf{Q}$, our goal is to design a transform and a quantizer that minimize:

$$\|\mathbf{Q}\mathbf{K}^\top - \mathbf{Q}\hat{\mathbf{K}}^\top\|_F^2 = \sum_{j=1}^M (\mathbf{k}_j - \hat{\mathbf{k}}_j) \mathbf{M}_q (\mathbf{k}_j - \hat{\mathbf{k}}_j)^\top, \quad (5)$$

for a given rate. Denoting $\|\mathbf{a}\|_{\mathbf{M}}^2 = \mathbf{a} \mathbf{M} \mathbf{a}^\top$, the right side of (5) can be rewritten as $\sum_j \|\mathbf{k}_j - \hat{\mathbf{k}}_j\|_{\mathbf{M}_q}^2$, i.e., an input-weighted quadratic distortion with *constant* sensitivity matrix $\mathbf{M}_q$ (Linder et al., 1999).

To understand how a $\mathbf{M}_q$ -based distortion affects the design, consider quantization based on companding. In the SQ case, this involves defining an invertible mapping $h(x)$, so that given $\mathcal{Q}_u$, a uniform SQ, $\hat{x} = h^{-1}(\mathcal{Q}_u(h(x)))$. For vectors, this can be generalized by selecting an invertible matrix $\mathbf{U}$, so that $h_U(\mathbf{x}) = h(\mathbf{x}\mathbf{U})$. Then, companding can be applied directly by using $\mathcal{Q}_u$ for each entry (e.g., entry-wise SQ) and defining $\hat{\mathbf{x}} = h_U^{-1}(\mathcal{Q}_u(h_U(\mathbf{x})))$. High-resolution quantization theory shows that we can minimize the $\mathbf{M}_q$ distortion in (5) via companding and an entry-wise SQ by selecting the optimal $h_U(\cdot)$, i.e., the one that minimizes (5), satisfying $h'_U(\mathbf{x})h'_U(\mathbf{x})^\top = c\mathbf{M}_q$, which, for linear companding characterized by a matrix $\mathbf{U}$ implies $\mathbf{U}\mathbf{U}^\top = c\mathbf{M}_q$ (Linder et al., 1999). Orthogonal transforms are suboptimal for the high-resolution regime unless $\mathbf{M}_q \propto \mathbf{I}$.

The condition $\mathbf{U}\mathbf{U}^\top = c\mathbf{M}_q$ determines the transform only up to an orthogonal factor: if $\mathbf{U}$ satisfies it, so does $\mathbf{U}\mathbf{W}$ for any orthogonal $\mathbf{W}$, since $\mathbf{U}\mathbf{W}\mathbf{W}^\top\mathbf{U}^\top = \mathbf{U}\mathbf{U}^\top$. We fix this factor by optimizing energy compaction, as the KLT does in (2), but under the $\mathbf{M}_q$ -weighted cost of (5) rather than the Euclidean one. Assume M tokens in the calibration set and let $\bar{\mathbf{k}} = 1/M \sum_j \mathbf{k}_j$; centering the keys has no cost to attention (Appendix F.1). We write $\mathbf{R}^\dagger = (\mathbf{R}^\top \mathbf{M}_q^{-1} \mathbf{R})^{-1} \mathbf{R}^\top \mathbf{M}_q^{-1}$ for the $\mathbf{M}_q$ -weighted pseudoinverse.

Theorem 1. *Let $\tilde{\mathbf{k}}_j = \mathbf{k}_j - \bar{\mathbf{k}}$, $\tilde{\mathbf{S}}_k = \sum_j \tilde{\mathbf{k}}_j^\top \tilde{\mathbf{k}}_j$, and let*

$$\mathbf{R}_{K,p}^* = \arg \min_{\mathbf{R} \in \mathbb{R}^{d \times p}} \sum_{j=1}^M \|\tilde{\mathbf{k}}_j - \tilde{\mathbf{k}}_j \mathbf{R} \mathbf{R}^\dagger\|_{\mathbf{M}_q}^2. \quad (6)$$

Then $\mathbf{R}_{K,p}^ = \mathbf{M}_q^{1/2} \mathbf{E}_{1:p}$, where $\mathbf{E}\mathbf{\Lambda}\mathbf{E}^\top$ is the eigendecomposition of $\mathbf{M}_q^{1/2} \tilde{\mathbf{S}}_k \mathbf{M}_q^{1/2}$.*

Proof: Appendix F.2. We construct $\mathbf{R}_K$ by computing the eigendecomposition of $\mathbf{M}_q^{1/2} \tilde{\mathbf{S}}_k \mathbf{M}_q^{1/2}$ and then multiplying by $\mathbf{M}_q^{1/2}$; $\mathbf{R}_{K,p}^*$ corresponds to the matrix truncated by keeping the eigenvectors corresponding to the dominant eigenvalues of $\mathbf{M}_q^{1/2} \tilde{\mathbf{S}}_k \mathbf{M}_q^{1/2}$, i.e., $\mathbf{R}_K = \mathbf{M}_q^{1/2} \mathbf{E}$. The centered key is encoded as $\mathbf{r}_j = (\mathbf{k}_j - \bar{\mathbf{k}}) \mathbf{R}_K$ and reconstructed as $\hat{\mathbf{k}}_j = \mathbf{r}_j \mathbf{R}_K^{-1} + \bar{\mathbf{k}}$. We have $\mathbf{R}_K \mathbf{R}_K^\top = \mathbf{M}_q$, as demanded by (Linder et al., 1999), and the transform satisfies a generalized Parseval relation (Girault et al., 2018):

Proposition 1. *Let $\mathbf{r}$ and $\hat{\mathbf{r}}$ be any two vectors, and $\mathbf{k} = \mathbf{r} \mathbf{R}_K^{-1}$ and $\hat{\mathbf{k}} = \hat{\mathbf{r}} \mathbf{R}_K^{-1}$. Then, $\|\mathbf{r} - \hat{\mathbf{r}}\|_2^2 = \|\mathbf{k} - \hat{\mathbf{k}}\|_{\mathbf{M}_q}^2$.*

Prop. 1 is analogous to the Parseval relation in Sec. 2.2 for the $\mathbf{M}_q$ -MSE: the transform converts the weighted objective into ordinary MSE. Thus, we can run any off-the-shelf MSE-optimal quantizer on the transform coefficients. Since $\mathbf{R}_K$ satisfies the linear compander condition, it is optimal for the associated high-resolution attention-weighted companding problem. Its orthogonal factor is chosen to optimize the weighted low-rank approximation in Theorem 1.

3.2 Value transform

We derive the value transform from $\|\mathbf{S}\mathbf{V} - \mathbf{S}\hat{\mathbf{V}}\|_F^2 = \sum_{j,j'} (\mathbf{M}_s)_{jj'} (\mathbf{v}_j - \hat{\mathbf{v}}_j)(\mathbf{v}_{j'} - \hat{\mathbf{v}}_{j'})^\top$, and $\mathbf{M}_s = \mathbf{S}^\top \mathbf{S}$, the second moment of the attention scores.

Corollary 1. *Let $\mathbf{o}_i = \mathbf{s}_i \mathbf{V}$ denote the attention outputs on calibration data, and $\mathbf{E}\mathbf{\Lambda}\mathbf{E}^\top$ the eigendecomposition of $\mathbf{M}_o = \mathbf{V}^\top \mathbf{M}_s \mathbf{V}$. The minimizer of $\|\mathbf{S}\mathbf{V} - \mathbf{S}\hat{\mathbf{V}}\|_F^2$ over low-rank approximations is $\mathbf{R}_V = \mathbf{E}_{1:p}$, with codes $\mathbf{s}_j = \mathbf{v}_j \mathbf{R}_V$ and reconstruction $\hat{\mathbf{v}}_j = \mathbf{s}_j \mathbf{R}_V^\top$.*

The proof is in Appendix F.4. Unlike $\mathbf{R}_K$, the value transform is orthogonal. The term $\mathbf{V}^\top \mathbf{M}_s \mathbf{V}$ is the covariance used in (Zhou et al., 2026).

4 KV cache vector quantization

Our goal is to design quantizers to minimize distortion at a fixed rate, because serving imposes a fixed-width constraint: tokens must occupy a fixed number of bits, and rate cannot be reallocated across entries. We

achieve this via VQ, in which groups of entries are jointly quantized. This leaves the choice of partition, i.e., how to group the entries, as the only degree of freedom. We show that under standard information-theoretic assumptions, there is a criterion that identifies which groupings are rate-distortion optimal when per-group rate is fixed, and we show how to find them.

Let π denote a partition of $\{1, \dots, d\}$ into $L = d/g$ groups $G_1, \dots, G_L$ of equal size g , and let $\mathbf{r}_{G_\ell}$ be the transform coefficients indexed by G_ℓ . We first assume that each group has its own codebook $\mathcal{C}_\ell$ with b_ℓ bits, subject to $\sum_\ell g b_\ell = db$. Later, we enforce the constraint that $b_\ell = b$. By Proposition 1, we can work with the MSE, which satisfies $D(\pi, \{b_\ell\}) = \sum_\ell D_\ell(b_\ell)$. We model the coefficients as independent zero-mean Gaussians with variances $\sigma_1^2 \geq \dots \geq \sigma_d^2$ (independence is assumed for tractability.) Under the high-resolution regime, the distortion-rate function of the optimal g -dimensional quantizer for group G_ℓ is (Zador, 1982):

$$D_\ell(b_\ell) = C_g 2^{-2b_\ell} \left(\prod_{i \in G_\ell} \sigma_i^2 \right)^{1/g} (1 + o(1)), \quad (7)$$

where C_g is common to all groups. The $o(1)$ term in (7) vanishes for high rates, and we assume continuous rates $b_\ell \in \mathbb{R}$. The partition enters (7) only through the *volume* $v_\ell(\pi) \doteq \prod_{i \in G_\ell} \sigma_i^2$. We seek the partition and per-group rates minimizing $D(\pi, \{b_\ell\})$, assuming: *variable rates*, where $\{b_\ell\}$ can take any value subject to the budget $\sum_\ell b_\ell = Lb$, and *fixed rates*, where $b_\ell \equiv b$ as required by fixed-width packing. Note that $\prod_\ell v_\ell(\pi) = \prod_{i=1}^d \sigma_i^2$ for every π .

Theorem 2. *Under the model (7), for any partition π the allocation minimizing $D(\pi, \{b_\ell\})$ subject to $\sum_\ell b_\ell = Lb$ is*

$$b_\ell^*(\pi) = b + \frac{1}{2g} \log_2 \left(v_\ell(\pi) / \prod_m v_m(\pi)^{1/L} \right), \quad (8)$$

which spends more bits on groups of larger volume. The optimal distortion, $D^(b) = C_g L 2^{-2b} (\prod_{i=1}^d \sigma_i^2)^{1/d}$, is the same for every π , assuming that rates can be the arbitrary real values from (8).*

The proof is in Appendix G.1. The allocation (8) depends on the partition only through the volumes, and it is uniform, $b_\ell^* = b$ for all ℓ , only when $v_1(\pi) = \dots = v_L(\pi)$. Hence, for a volume-equalizing partition, the optimal allocation is already fixed-width. Instead, if the volumes are not equal and we apply the same number of bits, $b_\ell \equiv b$, the distortion will be suboptimal, $2^{-2b} \sum_\ell v_\ell(\pi)^{1/g} \geq D^*(b)$, with equality iff $v_1(\pi) = \dots = v_L(\pi)$. We therefore group entries into equal-volume groups and spend the same number of bits per group; within each group, the codebook places more resolution on the entries of higher variance in the query-weighted transform domain.

The distortion of a partition is proportional to the average of the $v_\ell(\pi)^{1/g}$, while $D^*(b)$ is proportional to their geometric mean, so the penalty grows with the spread of the group volumes. Sorting the entries and grouping them consecutively maximizes spread, while a random partition averages log-variances per group (closer to balanced). Balancing $\sum_{i \in G_\ell} \log \sigma_i^2$ across groups is NP-hard (reduces to 3-Partition (Garey and Johnson, 2002)); we use a heuristic. Since each vector has g entries, we will have $L = d/g$ vectors to quantize. Then, we sort the entries by decreasing order of $\log \sigma_i^2$ and, to approximately equalize their volume, group G_ℓ contains entries with index $i \equiv \ell - 1 \pmod L$. The partition is folded into the transform, so it costs nothing at inference.

We apply VQ to the keys and keep SQ for the values since they are well approximated at this rate: replacing the value quantizer with VQ leaves accuracy almost unchanged (Appendix B). **Complexity.** Write, once per token: centering, and a nearest-neighbor search over centroids per group. Read: a lookup into a codebook of $2^{gb}g$ scalars per group, and one multiply to restore per-token scale.

5 Experiments

We use 12 A100 GPUs (40GB); for throughput, we use a single H100 (details in Appendix B). The high-resolution model might not hold at 2 bits, we use the theory to guide our design rather than as a guarantee. As models, we use Llama-3.1-8B (Grattafiori et al., 2024), Qwen3, both 8B and 4B-Thinking (Yang et al., 2025a),

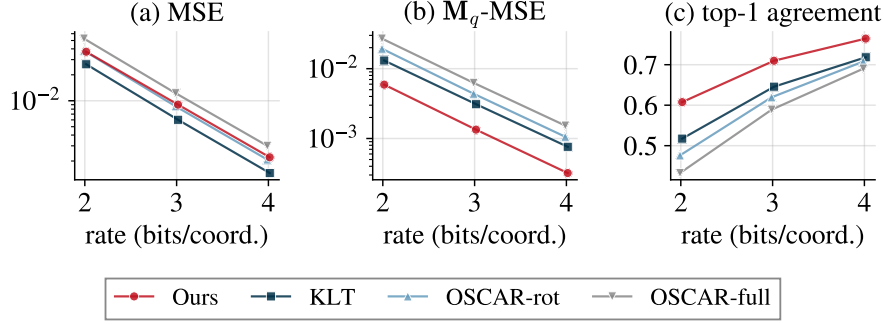


Figure 3 Bases under a fixed quantizer. The KLT minimizes MSE (a) but not top-1 agreement (c); the M_q -MSE (b) orders the bases like (c); our transform is best in (b-c).

and GPT-OSS (Agarwal et al., 2025). For GPT-OSS, we quantize the KV cache only in its full-attention layers, leaving sliding-window layers (128-token window) unchanged.

Transform analysis. We fix the quantizer: uniform SQ per coordinate with per-coordinate entropy coding, fit on the calibration split and evaluated on held-out data. Entropy coding removes rate allocation as a confound, so the comparison reflects the basis alone; OSCAR’s transforms target fixed-rate SQ and gain no compaction here. We compare four bases: the KLT of the keys (MSE-optimal), the eigenbasis of the query second moment used by OSCAR, the full OSCAR transform, and ours. Fig. 3 shows the MSE, the query-weighted distortion (5), and the top-1 attention agreement, i.e., the fraction of queries whose highest-scoring key survives quantization. MSE does not determine attention fidelity: the KLT attains the lowest MSE at every rate (a), yet ours, with a higher MSE, preserves the top-1 key more often (c).

Table 2 RULER NIAH mean accuracy (%) across context lengths, mean $\pm$ 95% CI over 3 evaluation rollouts. BPE is the effective bits per KV element at 128K context. Best compressed method per column in **bold**.

Model	Method	BPE	8K	16K	32K	64K	128K
Qwen3-4B -Thinking-2507	BF16	16	99.5 $\pm$ 0.4	98.4 $\pm$ 0.4	98.7 $\pm$ 0.2	93.1 $\pm$ 0.8	89.9 $\pm$ 2.1
	QuaRot	2.25	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	13.4 $\pm$ 0.8	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	OSCAR	2.28	99.7 $\pm$ 0.0	94.0 $\pm$ 0.7	81.6 $\pm$ 0.8	63.0 $\pm$ 0.8	34.7 $\pm$ 0.5
	NOVA-KV (ours)	2.22	99.2 $\pm$ 0.4	97.1 $\pm$ 0.0	97.8 $\pm$ 0.1	82.9 $\pm$ 0.8	79.2 $\pm$ 0.5
Qwen3-8B	BF16	16	99.9 $\pm$ 0.0	99.4 $\pm$ 0.0	98.5 $\pm$ 0.1	84.3 $\pm$ 0.6	83.4 $\pm$ 0.6
	QuaRot	2.25	54.8 $\pm$ 0.2	31.3 $\pm$ 0.2	23.5 $\pm$ 0.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	OSCAR	2.28	96.7 $\pm$ 0.3	94.6 $\pm$ 0.2	86.9 $\pm$ 0.2	60.6 $\pm$ 0.6	25.3 $\pm$ 0.8
	NOVA-KV (ours)	2.22	99.4 $\pm$ 0.1	98.7 $\pm$ 0.0	96.0 $\pm$ 0.2	76.4 $\pm$ 0.4	75.4 $\pm$ 0.1
Llama-3.1-8B	BF16	16	97.9 $\pm$ 3.0	98.3 $\pm$ 1.8	97.3 $\pm$ 3.3	97.3 $\pm$ 2.4	84.2 $\pm$ 10.7
	QuaRot	2.25	37.2 $\pm$ 25.2	33.8 $\pm$ 26.5	35.2 $\pm$ 26.6	34.7 $\pm$ 27.1	4.8 $\pm$ 5.5
	OSCAR	2.28	86.1 $\pm$ 12.2	81.6 $\pm$ 17.7	82.4 $\pm$ 19.4	76.9 $\pm$ 21.9	36.7 $\pm$ 25.2
	NOVA-KV (ours)	2.22	94.5 $\pm$ 3.4	91.9 $\pm$ 5.9	92.1 $\pm$ 7.2	92.4 $\pm$ 5.6	63.3 $\pm$ 24.6
GPT-OSS-20B	BF16	16	95.8 $\pm$ 4.4	95.7 $\pm$ 3.4	94.6 $\pm$ 3.2	92.2 $\pm$ 5.5	80.4 $\pm$ 2.2
	QuaRot	2.50	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	OSCAR	2.53	0.5 $\pm$ 0.2	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
	NOVA-KV (ours)	2.41	89.6 $\pm$ 1.1	81.0 $\pm$ 2.6	79.2 $\pm$ 3.8	70.6 $\pm$ 4.1	54.0 $\pm$ 4.4

Calibration. As in (Zhou et al., 2026), the key transform and codebooks are fit offline from 198 GPQA-Diamond prompts, run in prefill only, with per-(layer, KV-head) statistics. For GQA (Ainslie et al., 2023), we calibrate across heads sharing the same KV cache. For the keys, we fit codebooks with 256 centroids (vector size 4, average 2 bits/coordinate) by k -means. We concatenate these prompts into long-context sequences of 128K so the data spans the RoPE positions at which the transform and codebooks are evaluated. No label, gold answer, model generation, or task metric enters the fit. For values, we use the SQ from (Zhou et al., 2026). Refitting the transform, grouping, and codebooks on MMLU instead, a disjoint domain, leaves

Table 3 KV-cache quantization compared across models and benchmarks. Entries are mean $\pm$ 95% CI with 5 samples per prompt. BPE denotes effective bits per KV element at 128K context. “Drop” is the gap in the Mean column to the BF16 reference; less negative is better. Best compressed method per column in **bold**. Naive INT2 yields 0.0 for all models and tasks.

Model	Method	BPE	GPQA	HumanE	LCB v6	AIME25	MATH500	Mean	Drop
Qwen3-4B -Thinking-2507	BF16	16.00	64.9 $\pm$ 0.8	84.8 $\pm$ 0.5	59.2 $\pm$ 0.9	70.0 $\pm$ 1.5	97.5 $\pm$ 0.2	75.3	–
	TurboQuant	3.25	44.3 $\pm$ 1.5	31.2 $\pm$ 0.4	1.3 $\pm$ 1.5	2.2 $\pm$ 3.8	56.6 $\pm$ 0.7	27.1	–48.2
	QuaRot	2.25	12.6 $\pm$ 1.7	6.1 $\pm$ 1.3	3.0 $\pm$ 1.2	0.0 $\pm$ 0.0	32.5 $\pm$ 1.3	10.8	–64.5
	OSCAR	2.28	62.4 $\pm$ 0.9	83.8 $\pm$ 0.2	57.4 $\pm$ 0.8	63.3 $\pm$ 2.1	96.8 $\pm$ 0.2	72.7	–2.6
	NOVA-KV (ours)	2.22	62.9 $\pm$ 1.3	84.3 $\pm$ 0.4	56.6 $\pm$ 0.6	67.3 $\pm$ 1.2	97.4 $\pm$ 0.2	73.7	–1.6
Qwen3-8B	BF16	16.00	57.2 $\pm$ 0.9	91.7 $\pm$ 0.5	57.6 $\pm$ 0.7	70.0 $\pm$ 1.7	97.0 $\pm$ 0.1	74.7	–
	TurboQuant	3.25	45.6 $\pm$ 6.2	76.0 $\pm$ 1.9	32.7 $\pm$ 2.5	45.6 $\pm$ 5.1	94.3 $\pm$ 0.1	58.8	–16.9
	QuaRot	2.25	44.2 $\pm$ 1.3	39.0 $\pm$ 3.7	9.0 $\pm$ 2.0	20.0 $\pm$ 3.3	75.2 $\pm$ 0.4	37.5	–37.2
	OSCAR	2.28	53.5 $\pm$ 1.2	90.9 $\pm$ 0.7	53.6 $\pm$ 0.2	66.2 $\pm$ 1.7	96.7 $\pm$ 0.2	72.2	–2.5
	NOVA-KV (ours)	2.22	55.1 $\pm$ 1.5	92.1 $\pm$ 0.2	55.6 $\pm$ 0.6	64.6 $\pm$ 2.5	96.9 $\pm$ 0.2	72.9	–1.8
Llama-3.1-8B	BF16	16.00	25.5 $\pm$ 2.7	63.5 $\pm$ 2.3	10.5 $\pm$ 1.7	0.0 $\pm$ 0.0	44.2 $\pm$ 1.9	28.7	–
	TurboQuant	3.25	22.7 $\pm$ 2.3	56.7 $\pm$ 2.8	11.2 $\pm$ 1.7	0.0 $\pm$ 0.0	37.3 $\pm$ 1.2	25.6	–3.2
	QuaRot	2.25	2.8 $\pm$ 1.0	6.7 $\pm$ 1.4	0.7 $\pm$ 0.5	0.7 $\pm$ 1.3	6.0 $\pm$ 0.8	3.4	–25.4
	OSCAR	2.28	23.1 $\pm$ 2.6	64.4 $\pm$ 2.5	9.4 $\pm$ 1.4	3.3 $\pm$ 4.1	44.8 $\pm$ 1.1	29.0	+0.3
	NOVA-KV (ours)	2.22	21.3 $\pm$ 2.6	64.5 $\pm$ 1.6	10.2 $\pm$ 1.7	3.3 $\pm$ 2.9	45.8 $\pm$ 1.3	29.0	+0.3
GPT-OSS-20B	BF16	16.00	50.4 $\pm$ 3.1	90.2 $\pm$ 2.9	76.2 $\pm$ 2.3	74.0 $\pm$ 7.0	91.8 $\pm$ 1.1	76.5	–
	TurboQuant	3.50	35.2 $\pm$ 3.0	33.4 $\pm$ 3.8	30.2 $\pm$ 4.1	17.3 $\pm$ 3.5	70.9 $\pm$ 1.8	37.4	–39.1
	QuaRot	2.50	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0	–76.5
	OSCAR	2.53	25.1 $\pm$ 2.7	5.4 $\pm$ 1.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	39.0 $\pm$ 1.9	13.9	–62.6
	NOVA-KV (ours)	2.41	46.5 $\pm$ 3.1	84.6 $\pm$ 3.6	71.9 $\pm$ 2.5	67.3 $\pm$ 7.5	91.8 $\pm$ 1.1	72.4	–4.1

RULER NIAH within 2.0 points and does not lower GPQA, so calibrating on the evaluation domain confers no advantage (Appendix C).

Baselines. As in (Xiao et al., 2024; Zhou et al., 2026), we keep the first 64 tokens and the most recent 256 in BF16 and quantize the rest. We run QuaRot, TurboQuant, and OSCAR under the configuration of (Zhou et al., 2026): matching group sizes, per-channel/per-token axes, and BF16 sink-plus-recent band. For OSCAR we evaluated both the released transform and one refit on our calibration split; the two are within 1.5 points on RULER-NIAH, and we report the better of the two. TurboQuant is included at 3.25 BPE, following (Zhou et al., 2026); Zandieh et al. (2025) report quality neutrality at 3.5 bits per channel and marginal quality degradation at 2.5.

Long context. We use RULER NIAH (Hsieh et al., 2024), a benchmark for long context robustness, and sweep the context from 8K to 128K. We run under chunked prefill, as serving engines do to bound iteration latency: the prompt is processed in fixed-size chunks and each chunk is quantized on write, so every subsequent chunk attends to an already quantized history. Deferring quantization until the whole prompt is prefilled avoids this accumulation, but holds the cache in BF16 at its peak. All methods run in the same harness with the same chunk size. Numbers for prior methods might be lower than those reported in the original papers, which do not specify a chunked prefill. Yet, our implementations remain comparable to their reported behavior on the reasoning and coding benchmarks (cf. Table 3). We report the mean over 3 rollouts with a 95% CI in Table 2. The baselines degrade with context while NOVA-KV tracks BF16 more closely: on Qwen3-8B at 128K, QuaRot and OSCAR reach 0.0 and 25.3 against 83.4 for BF16, where NOVA-KV reaches 75.4. At 128K, NOVA-KV exceeds OSCAR by 50.1 points (McNemar $p < 10^{-37}$). The same ordering holds on Llama-3.1-8B, where NOVA-KV reaches 92.4 at 64K and 63.3 at 128K against 76.9 and 36.7 for OSCAR.

The separation is starkest on GPT-OSS-20B, where both baselines fail rather than degrading with context: QuaRot scores 0.0 at every length, and OSCAR reaches 0.5 at 8K and 0.0 from 16K on, so neither retrieves anything at any context we test. NOVA-KV instead retains 89.6 at 8K and 54.0 at 128K, against 95.8 and 80.4 for BF16. This is the only model on which the orthogonal-transform baselines lose the task entirely, and it is also the one whose architecture is closest to current practice: GPT-OSS-20B interleaves sliding-window and full-attention layers and routes its feed-forward blocks through a mixture of experts, a combination that recent open-weight models have increasingly adopted.

Accuracy comparisons. We test on GPQA-Diamond (Rein et al., 2023), HumanEval (Chen et al., 2021), LiveCodeBench v6 (Jain et al., 2025), AIME25 (Mathematical Association of America, 2025), and MATH500 (Lightman et al., 2024; Hendrycks et al., 2021). We draw 5 samples per prompt (Table 3). Against BF16, NOVA-KV shows no detectable degradation on either Qwen model (McNemar $p = 0.17, 0.24$), while OSCAR is significantly below BF16 on Qwen3-4B ($p = 0.002$). The two compressed methods are not separable there ($p = 0.46$), and on these two models the significant separation appears only on long-context retrieval. The remaining two models separate them directly. On Llama-3.1-8B every arm tracks BF16 within the confidence intervals, so the methods are hard to tell apart: the uncompressed model already scores low on these tasks. On GPT-OSS-20B, by contrast, NOVA-KV preserves most of the BF16 accuracy on every benchmark (mean 72.4 against 76.5), whereas OSCAR collapses, scoring 5.4 on HumanEval and 0.0 on both LiveCodeBench v6 and AIME25 for a mean of 13.9, and TurboQuant falls to 37.4. NOVA-KV therefore stays close to BF16 on all four models, while the robustness of the orthogonal-transform baselines varies considerably across architectures.

Bit rate. At 128K context, the K cache stores one 8 bit index per group of $g = 4$ entries (2 BPE, as $|C|=256$) together with a per-token BF16 scale ($16/128 = 0.125$). The V cache is INT2 with a BF16 scale and offset (zero-point), giving $2.000 + 32/128 = 2.250$. Averaging over the 256 elements gives 2.1875, and the BF16 sink-plus-recent band (64, 256) adds 0.034, for 2.22 BPE. OSCAR stores a zero-point on both caches, giving 2.28 BPE; the 0.06 gap is the K-side zero-point. Transforms and codebooks are stored per model.

Throughput. We measure decode throughput in SGLang (Zheng et al., 2024) at input lengths of 30K, 60K and 90K (Fig. 4). We include a warm-up request, and throughput is computed over the decode window alone, excluding time to first token. Cross-request prefix sharing is disabled. We report aggregate tokens per second across the batch. At batch 1 and batch 4 the comparison is bandwidth-dominated: NOVA-KV reaches $1.6\text{--}3.1\times$ BF16 on Qwen3-8B and $1.7\text{--}3.4\times$ on Qwen3-4B, with the larger factors at the longer inputs. The factor is smaller on GPT-OSS-20B, $1.1\text{--}1.5\times$, since sliding-window reduces the share of the cache being compressed. Against OSCAR at the same rate, the two arms are within a few percent of each other throughout. At 90K the BF16 KV pool admits at most **b4** on either Qwen model, so it has no batch-16 configuration, whereas both 2-bit arms serve one.

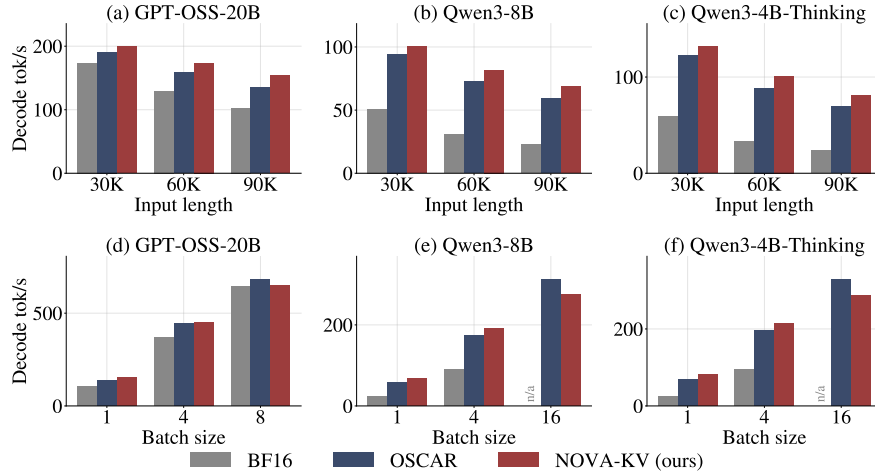


Figure 4 Decode throughput, prefill excluded. Top row: batch 1 against input length. Bottom row: batch scaling at 90K input. BF16+Qwen does not admit the largest batch at 90K.

Transform-quantizer. Table 4 crosses transform (OSCAR, NOVA-KV) and quantizer. Under SQ (same number of bits to each coordinate), our transform collapses: it compacts energy rather than flattening it, so a single rate cannot serve every coordinate. Replacing SQ by VQ is worth 7.5 and 37.4 points under OSCAR’s basis; replacing OSCAR’s basis by ours, under VQ, is worth a further 8.3 and 12.7. Neither substitution alone reaches the full result.

Table 4 Transform-quantizer ablation on RULER NIAH, Qwen3-8B. Mean accuracy (%) $\pm$ 95% CI over 3 rollouts.

Transform	Quantizer	BPE	64K	128K
OSCAR	SQ	2.28	60.6 $\pm$ 0.6	25.3 $\pm$ 0.8
OSCAR	VQ	2.22	68.1 $\pm$ 0.6	62.7 $\pm$ 0.4
NOVA-KV	SQ	2.28	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
NOVA-KV	VQ	2.22	76.4 $\pm$ 0.4	75.4 $\pm$ 0.1

Volume equalization. We fix the transform, the codebook size, and the rate (2.22 BPE), varying only the partition: *variance-sorted* groups consecutive entries in decreasing variance, the configuration with the most unbalanced volumes; *random* draws one fixed partition at random; and *equalizing* is the partition of Sec. 4. Over 3 evaluation rollouts for Qwen3-8B, variance-sorted is the worst (53.2 $\pm$ 0.7 at 64K, 37.2 $\pm$ 0.1 at 128K), random recovers most of the gap (74.2 $\pm$ 0.5, 74.4 $\pm$ 0.2), and the equalizing partition is the highest (76.4 $\pm$ 0.4, 75.4 $\pm$ 0.1), matching Theorem 2.

6 Conclusion

We formulated KV cache quantization as a transform coding problem whose distortion is the attention product error. Under a high-resolution model, the optimal key transform is not orthogonal, and satisfies a generalized Parseval relation, turning the attention-aware criterion into MSE in the transform domain. Under the independent-Gaussian high-resolution model, equal-volume grouping makes fixed-rate VQ attain the variable-rate optimum, allowing a fixed-width cache without an asymptotic distortion penalty. At two bits per element, NOVA-KV reduces the gap to BF16 relative to the 2-bit state-of-the-art at comparable decoding speed. Joint quantization of keys and values is left for future work.

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A Notation and Definitions

This appendix collects the notation and definitions used throughout the paper.

A.1 Conventions

Uppercase bold letters ($\mathbf{A}$) denote matrices, lowercase bold letters ($\mathbf{a}$) denote row vectors, and regular letters denote scalars. The n th entry of $\mathbf{a}$ is a_n and the (i, j) th entry of $\mathbf{A}$ is A_{ij} . All vectors are row vectors, so a linear map $\mathbf{R}$ acts on the right: $\mathbf{xR}$. Eigenvalues are indexed in decreasing order, $\lambda_1 \geq \dots \geq \lambda_d$, and $\mathbf{E}_{1:p}$ collects the eigenvectors associated with the p largest eigenvalues. For a symmetric positive definite (p.d.) $\mathbf{M}$ we let $\|\mathbf{a}\|_{\mathbf{M}}^2 = \mathbf{aMa}^\top$, and $\mathbf{M}^{1/2}$ denotes the symmetric square root, i.e., the unique symmetric p.d. $\mathbf{B}$ with $\mathbf{BB} = \mathbf{M}$. We write $\|\cdot\|_F$ for the Frobenius norm and $\|\cdot\|_2$ for the spectral norm of a matrix and the ℓ_2 norm of a vector.

A.2 Attention and the cache

Notation for attention, RoPE, and the cache follows Sec. 2.1 of the main text. We let $\hat{\mathbf{K}}, \hat{\mathbf{V}}$ be the reconstructed keys and values after quantization, and $\hat{\mathbf{S}}$ be the scores computed from $\hat{\mathbf{K}}$. Under grouped-query attention (GQA) (Ainslie et al., 2023), the query heads are partitioned into groups, and each group shares a single key-value projection pair $(\mathbf{W}_K, \mathbf{W}_V)$; we call the shared projection pair, together with the cache entries it produces, a *KV head*. Quantities are then formed per (layer, KV head); the query statistic $\mathbf{M}_q$ is accumulated over the query heads sharing the KV head (Appendix F.5), and the transform, grouping, and codebooks are fit per KV head.

A.3 Calibration statistics

Let M be the number of cached keys collected on calibration. The calibration statistics are the key mean $\bar{\mathbf{k}} = 1/M \sum_j \mathbf{k}_j$, the key covariance $\tilde{\mathbf{S}}_k = \sum_j (\mathbf{k}_j - \bar{\mathbf{k}})^\top (\mathbf{k}_j - \bar{\mathbf{k}})$, the query second moment $\mathbf{M}_q = \mathbf{Q}^\top \mathbf{Q}$, the score second moment $\mathbf{M}_s = \mathbf{S}^\top \mathbf{S}$, and the output second moment $\mathbf{M}_o = \mathbf{V}^\top \mathbf{M}_s \mathbf{V}$. $\mathbf{M}_q$ is positive semidefinite by construction; we assume it is positive definite for the analysis. Numerically, $\mathbf{M}_q^{1/2}$ and $\mathbf{M}_q^{-1/2}$ are formed from its symmetric eigendecomposition after flooring the eigenvalues at 10^{-30} ; no additive ridge is applied. For values, we use the default rotation matrix provided in (Zhou et al., 2026).

A.4 Transforms, quantization, and grouping

For the keys, the transform is $\mathbf{R}_K = \mathbf{M}_q^{1/2} \mathbf{E}$, with $\mathbf{E}$ the eigenvectors of $\mathbf{M}_q^{1/2} \tilde{\mathbf{S}}_k \mathbf{M}_q^{1/2}$ (Theorem 1); the transform coefficients of key j are $\mathbf{r}_j = (\mathbf{k}_j - \bar{\mathbf{k}}) \mathbf{R}_K$, and $\sigma_1^2 \geq \dots \geq \sigma_d^2$ denote their per-coordinate variances. The query-weighted distortion, or $\mathbf{M}_q$ -MSE, of a reconstruction is $\|\mathbf{k} - \hat{\mathbf{k}}\|_{\mathbf{M}_q}^2$; summed over cached keys it equals the key-query inner-product error (cf. Eq. (10)).

A quantizer of group size g maps subvectors of g entries jointly; a partition π of $\{1, \dots, d\}$ into $L = d/g$ groups $G_1, \dots, G_L$ of size g has group *volume* $v_\ell(\pi) = \prod_{i \in G_\ell} \sigma_i^2$, and is *volume-equalizing* when $v_1(\pi) = \dots = v_L(\pi)$. Under *fixed rates* every group gets the same rate $b_\ell \equiv b$; under *variable rates* $\{b_\ell\}$ is free subject to $\sum_\ell b_\ell = Lb$.

In (7), we let $C_g = g Z_g \gamma_g$, with Z_g the normalized moment of inertia of the optimal g -dimensional cell shape and γ_g the source-density functional of Zador’s theorem; under the independent-Gaussian model C_g is common to all groups (Zador, 1982) (no effect on the choice of partition).

The groupings compared in this paper are: *variance-sorted*, $G_\ell = \{(\ell - 1)g + 1, \dots, \ell g\}$ after sorting entries by decreasing σ_i^2 ; *random*, one partition drawn uniformly at random and then held fixed; and *equalizing*, the partition of Sec. 4, $G_\ell = \{i \in \{1, \dots, d\} : i - 1 \equiv \ell - 1 \pmod{L}\}$ after the same sort. Fig. 5 illustrates the three rules.

A.5 Symbol table

We summarize important notations in Table 5.

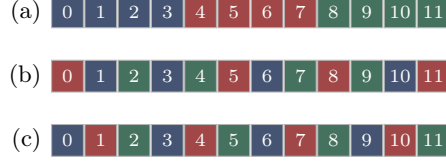


Figure 5 The three partitions in a zero-indexed toy example ($d = 12$, $g = 4$, $L = 3$); color denotes group membership, entries ordered by decreasing variance σ_i^2 from left to right. (a) *Variance-sorted* groups consecutive spectrum entries. (b) *Random* mixes levels by chance. (c) *Equalizing* deals coordinate i to group $i \bmod L$, so each group receives one coordinate per level (Theorem 2).

Table 5 Summary of notations used in this work.

Symbol	Meaning
d	head dimension
T	sequence length
M	number of calibration keys
$\mathbf{q}_t, \mathbf{k}_t$	post-RoPE query and key at position t
$\mathbf{v}_t$	value at position t
$\mathbf{S}, \hat{\mathbf{S}}$	attention scores from $\mathbf{K}$, from $\hat{\mathbf{K}}$
$\bar{\mathbf{k}}$	calibration key mean
$\mathbf{M}_q, \tilde{\mathbf{S}}_k, \mathbf{M}_s, \mathbf{M}_o$	calibration second-order statistics
$\mathbf{R}_K, \mathbf{R}_V$	key and value transforms
$\mathbf{r}_j$	transform coefficients of key j
σ_i^2	variance of transform coordinate i
$\mathcal{Q}^+, \mathcal{Q}^-, \mathcal{Q}$	encoder, decoder, their composition
$\mathcal{C}_\ell$	codebook of group ℓ
g, L	group size, number of groups
b, b_ℓ	rate, per-group rate
$\pi, G_\ell, v_\ell(\pi)$	partition, group, group volume
$n_{\text{sink}}, n_{\text{rec}}$	sink and recent band sizes
SQ, VQ	scalar quantization, vector quantization

Algorithm 1 Query-aware key transform and quantizer

Calibration (offline):

- 1: Collect queries $\mathbf{Q}$ and keys $\mathbf{K}$ on calibration data
- 2: $\bar{\mathbf{k}} \leftarrow \sum_{j=1}^M \mathbf{k}_j / M$; $\mathbf{M}_q \leftarrow \mathbf{Q}^\top \mathbf{Q}$; $\tilde{\mathbf{S}}_k \leftarrow \sum_{j=1}^M (\mathbf{k}_j - \bar{\mathbf{k}})^\top (\mathbf{k}_j - \bar{\mathbf{k}})$
- 3: Eigendecompose $\mathbf{M}_q^{1/2} \tilde{\mathbf{S}}_k \mathbf{M}_q^{1/2} = \mathbf{E} \mathbf{\Lambda} \mathbf{E}^\top$
- 4: $\mathbf{R}_K \leftarrow \mathbf{M}_q^{1/2} \mathbf{E}$; $\mathbf{R}_K^{-\top} = \mathbf{M}_q^{-1/2} \mathbf{E}$
- 5: Find volume-equalized groups π (Thm. 2), fold into $\mathbf{R}_K$
- 6: Train per-group codebooks $\{\mathcal{C}_\ell\}$ on $(\mathbf{K} - \mathbf{1}\bar{\mathbf{k}})\mathbf{R}_K$

Inference (online):

- 7: Write: cache $\hat{\mathbf{K}} = \mathcal{Q}^+((\mathbf{k}_j - \bar{\mathbf{k}})\mathbf{R}_K)$
 - 8: Attention: $\mathbf{S} = \text{softmax}(\mathbf{Q} \mathbf{R}_K^{-\top} \mathcal{Q}^-(\hat{\mathbf{K}})^\top / \sqrt{d})$
-

A.6 High-level algorithm

In Algorithm 1, we detail our pipeline both during calibration and inference.

B Experimental Details

In this section, we detail our experimental setup, including metrics, decoding configuration, calibration, serving setup, benchmarks, and datasets. We also ablate the VQ and SQ in the value arm to justify our deployed method.

B.1 Reported quantities

Bits per element (BPE). The average number of stored bits per cached scalar entry, over keys and values and accounting for the full context, including per-token scales, zero-points, and the BF16 sink-plus-recent band, and excluding the per-model transforms and codebooks, whose contribution is reported separately. The cache-rate accounting (2.22 for ours, 2.28 for OSCAR) is given in the main text.

The per-model transforms and codebooks are stored once and are independent of context and batch. Per (layer, KV head) we store the three $d \times d$ transforms $\mathbf{R}_K, \mathbf{R}_K^{-\top}, \mathbf{R}_V$, the mean $\bar{\mathbf{k}}$ (d scalars), and $L = d/g$ key codebooks of 2^{g_b} centroids in $\mathbb{R}^g$, that is $3d^2 + d + L 2^{g_b}g$ scalars. For instance, with Qwen3-8B, for $d = 128$, $g = 4$, $2^{g_b} = 256$, $L = 32$ this is $49152 + 128 + 32768 \approx 8.2 \times 10^4$ scalars per (layer, KV head); at 16-bit transforms and 8-bit codebooks it totals ≈ 0.30 Gbit for the full 36-layer, 8-KV-head model. Divided by the $2n_{\text{layer}}n_{\text{kv}}dL_{\text{ctx}}B$ quantized cache scalars, a $L_{\text{ctx}} = 128\text{K}$ sequence carries ≈ 0.03 bits per KV element; since the same transforms and codebooks serve the entire batch and all requests, the amortized contribution falls below 0.01 bits per KV element at the batch sizes we serve, negligible against the ≈ 2.2 cache rate.

Top-1 attention agreement. The fraction of queries whose highest-scoring key under $\hat{\mathbf{K}}$ is the same as under $\mathbf{K}$. It measures whether quantization preserves the argmax of the logits, which the MSE does not constrain directly.

Sink-plus-recent band. The first n_{sink} and the most recent n_{rec} tokens are held in BF16 and excluded from quantization; n_{sink} and n_{rec} are fixed constants, so the band’s share of the cache decreases as the context grows (see Fig. 6).

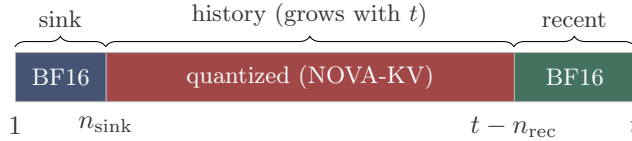


Figure 6 Cache layout at decoding position t . The first n_{sink} tokens (attention sinks) and the most recent n_{rec} tokens are held in BF16 and excluded from quantization; the tokens in between are stored in the NOVA-KV format. n_{sink} and n_{rec} are fixed ($n_{\text{sink}}=64$, $n_{\text{rec}}=256$), so the BF16 band’s share of the cache vanishes as the context grows.

Benchmarks and scoring. Table 10 lists, for each benchmark, the number of items scored (# items), the number of samples drawn per prompt (Smp.), the generation cap in tokens (Gen), and the per-item metric. In the main paper, the five generation benchmarks draw 5 samples per prompt under a 32K cap, while RULER NIAH uses 3 samples with a 128-token answer cap. The per-item metrics are as follows. *RULER NIAH*: case-insensitive substring match, with partial credit equal to the fraction of gold needles found. *GPQA-Diamond*: the answer letter is extracted with the simple-evals multiple-choice template and scored by exact match against the gold option (A–D). *HumanEval* and *LiveCodeBench v6*: pass@1, i.e., a sample is accepted only if it passes all reference unit tests under sandboxed execution. *AIME25* and *MATH500*: symbolic-equivalence checking of the final answer with the `math-verify` library. Benchmark accuracy is the mean of this per-item score over samples, reported with a bootstrap 95% confidence interval.

B.2 Decode-time breakdown (Fig. 1)

Figure 1 reports measured GPU kernel time per decoding step for Qwen3-8B served with SGLang on $8 \times \text{H100}$ 80 GB SXM GPUs using tensor parallelism of degree 8. We use a fixed context length of 16384 and a BF16

KV cache. Profiling covers decode steps 6–8, after the complete 16384-token prompt has been written to the cache, so every profiled step reads the full cache. Decode runs under CUDA graphs. Each bar is the median over the three steps of the summed CUDA-profiler kernel durations; CPU launch overhead and gaps between kernels are excluded.

We assign kernels to the five categories shown in Fig. 1 according to their names. *Parameter load* contains the linear and MLP GEMM kernels, whose time is dominated by accessing the model parameters in this regime. *Activations* contains the remaining normalization, elementwise, and activation-processing kernels. *KV load* contains the fused decode-attention kernels, which read the cached keys and values and compute the query-key products, softmax, and score-value products. *KV store* contains the cache-write kernels, and *all-reduce* contains the NCCL collectives.

The model parameter footprint is independent of batch size: the model has 15.1 GB of parameters in aggregate, and the same parameters are accessed once per decoding step regardless of the number of sequences in flight. Hence, the parameter-load bar remains nearly constant across batch sizes. KV-cache traffic, in contrast, scales with both batch size and context length. Each token contributes 512 bytes per KV head in BF16, including its key and value. Under tensor parallelism of degree 8, the per-GPU cache footprint therefore grows from 0.30 GB at batch 1 to 19.33 GB at batch 64. Correspondingly, the KV-load bar in Fig. 1 increases from 2.2 to 9.4 ms, or from 51% to 76% of the kernel time.

The KV-load category measures the duration of the fused attention kernels rather than memory traffic in isolation. These kernels stream the KV cache while performing the attention arithmetic. Accounting for reuse across grouped-query heads, decode attention has an arithmetic intensity of approximately 4.0 FLOP/byte, far below the H100 BF16 roofline ridge point of approximately 295 FLOP/byte. The fused attention kernel is therefore bandwidth-bound once sufficient parallelism is available. At batch 1, its measured KV traffic corresponds to only approximately 4% of the 3.35 TB/s peak HBM bandwidth because a single 16K-token sequence provides insufficient parallelism to occupy all 132 SMs. At batch 64, the achieved bandwidth rises to approximately 61% of peak. Thus, at large batch sizes, streaming the KV cache dominates the measured GPU kernel time.

B.3 Calibration

Statistics are accumulated over the calibration prompts, per (layer, KV head), in prefill only. The matrices $\mathbf{M}_q^{\pm 1/2}$ are formed from the symmetric eigendecomposition of $\mathbf{M}_q$ with eigenvalues floored at 10^{-30} before the $\pm 1/2$ power; no additive ridge is applied. The bases $\mathbf{R}_K$, $\mathbf{R}_K^{-\top}$, $\mathbf{R}_V$ are stored in float64 (the mean $\bar{\mathbf{k}}$ in float32) and cast to bfloat16 when loaded into the serving engine.

Key codebooks are trained by k -means with random initialization, for at most 25 iterations, on the 131072 transformed key subvectors per (layer, KV head) (8 calibration sequences $\times$ 16384 averaged positions) after centering, transformation, and grouping; empty clusters are reseeded each iteration from the points farthest from their centroid. There is one codebook per group per (layer, KV head), 256 centroids at $g = 4$, stored in float16 and snapped to fp8 offline; each cached subvector is an 8-bit index. Encoding is an exact nearest-neighbor search in the Euclidean metric of the transform domain. The grouping sorts entries by $\log \sigma_i^2$, taking σ_i^2 from the calibration spectrum, and deals them to the L groups in turn; the procedure is deterministic. We detail the algorithmic training steps in Algorithms 2 and 3.

The per-token scale ρ_j in Algorithm 3 normalizes each transformed key before the codebooks are trained. It is not part of the model of Theorem 2, which treats the entries as independent Gaussians with fixed variances σ_i^2 .

B.4 Serving and evaluation

We run in a research fork based on SGLang v0.5.10 with custom decoding kernels. Software versions for the two hardware setups are listed in Table 9. The accuracy evaluations use the Triton attention backend for both prefill and decode. Chunked prefill uses 4096-token chunks on the dense models and 8192 on GPT-OSS. RULER NIAH uses 8 subtasks (`niah_single_1/2/3`, `niah_multikey_1/2/3`, `niah_multivalue`, `niah_multiquery`); item counts, samples per prompt, generation caps, and per-item metrics are collected in

Algorithm 2 Query-aware transform and grouping

Input: M token positions; at each position j , the post-RoPE key $\mathbf{k}_j$ of this KV head and the post-RoPE queries, shared across H heads, $\mathbf{q}_j^{(1)}, \dots, \mathbf{q}_j^{(H)}$

Output: $\mathbf{R}_K, \mathbf{R}_K^{-\top}, \bar{\mathbf{k}}$

- 1: $\mathbf{M}_q \leftarrow \sum_{j=1}^M \sum_{h=1}^H \mathbf{q}_j^{(h)\top} \mathbf{q}_j^{(h)}$ $\triangleright$ summed over the H shared query heads (Appendix F.5)
 - 2: $\bar{\mathbf{k}} \leftarrow 1/M \sum_j \mathbf{k}_j$; $\tilde{\mathbf{S}}_k \leftarrow \sum_j (\mathbf{k}_j - \bar{\mathbf{k}})^\top (\mathbf{k}_j - \bar{\mathbf{k}})$
 - 3: $\mathbf{E} \mathbf{A} \mathbf{E}^\top \leftarrow \text{eig}(\mathbf{M}_q^{1/2} \tilde{\mathbf{S}}_k \mathbf{M}_q^{1/2})$, $\lambda_1 \geq \dots \geq \lambda_d$
 - 4: $\mathbf{R}_K \leftarrow \mathbf{M}_q^{1/2} \mathbf{E}$; $\mathbf{R}_K^{-\top} \leftarrow \mathbf{M}_q^{-1/2} \mathbf{E}$
 - 5: $\pi(r) \leftarrow r \bmod L$, $r = 0, \dots, d-1$ by decreasing λ
 - 6: $\mathbf{R}_K \leftarrow \mathbf{R}_K[:, \pi]$; $\mathbf{R}_K^{-\top} \leftarrow \mathbf{R}_K^{-\top}[:, \pi]$ $\triangleright$ folded in
-

Algorithm 3 Codebook training (per layer, KV head)

Input: keys $\{\mathbf{k}_j\}$, $\mathbf{R}_K, \bar{\mathbf{k}}$; rate b

Output: codebooks $\{\mathcal{C}_\ell\}_{\ell=1}^L$

- 1: **for** $j = 1, \dots, M$ **do**
 - 2: $\mathbf{r}_j \leftarrow (\mathbf{k}_j - \bar{\mathbf{k}}) \mathbf{R}_K$
 - 3: $\rho_j \leftarrow \|\mathbf{r}_j\|_2 / \sqrt{d}$; $\mathbf{r}_j \leftarrow \rho_j^{-1} \mathbf{r}_j$ $\triangleright$ Normalize
 - 4: **end for**
 - 5: **for** $\ell = 1, \dots, L$ **do**
 - 6: $\mathcal{C}_\ell \leftarrow k\text{-means}(\{\mathbf{r}_j[G_\ell]\}_j, 2^{gb})$ $\triangleright$ equal size for every group
 - 7: **end for**
-

Table 10. Baselines use their own reference calibration procedure, under the same sink-plus-recent band and the same chunk size as ours. *Naive INT2* denotes per-token INT2 applied directly to keys and values with no transform and no sink/recent band; it collapses to 0.0 on every model and task (main-text Table 3). For the OSCAR-basis rows of the transform-quantizer ablation (main-text Table 4), codebooks are refit on that basis with the same group size, codebook size, and training budget as ours; the KLT comparison (main-text Fig. 3) instead uses entropy-coded scalar quantization.

B.5 Decoding kernel and read path

The read path is a fused Triton kernel that never multiplies by the transform. The synthesis $\mathbf{R}_K^{-\top}$ is folded into the query once per step: the engine forms the modified query $\tilde{\mathbf{q}} = \mathbf{q} \mathbf{R}_K^{-\top}$ (a single $d \times d$ product, amortized over the whole cache), so scoring a cached key reduces to the inner product of $\tilde{\mathbf{q}}$ with the dequantized code $\mathcal{Q}^-(\bar{\mathbf{k}})$. For each of the L groups the stored 8-bit index selects a g -dimensional centroid from that group’s codebook; the kernel accumulates the partial logit across groups and applies the per-token scale ρ_j recovered from metadata (one multiply). Because a shared mean shifts every logit of a query by a constant, $\bar{\mathbf{k}}$ cancels in the softmax (Sec. F.1) and is not re-added on the read path. On the value side the INT2 code is dequantized in the OSCAR rotation basis, whose inverse is absorbed into the output projection, so it adds no online matmul either. Thus the per-element read cost is a table lookup and a scale, matching scalar dequantization. We implement the read path as a two-stage split-KV decode kernel, with gather and dequantize fused with the attention math in stage 1 and stage-2 split-combine; the split count follows the engine’s occupancy heuristic.

Value quantizer. We keep SQ for the values: replacing the value SQ with VQ barely changes downstream accuracy. We compared two configurations, the decorrelating transform $\mathbf{U}_S$ alone (as suggested by Cor. 1) and $\mathbf{U}_S$ followed by a Hadamard transform (closest to the deployed SQ value path). Both stay within 2 points of the deployed SQ (Table 6). While VQ is slightly more accurate at a slightly lower rate (2.16 vs. 2.22 BPE), we keep SQ for read-path simplicity.

Implementation We consider two implementations of the read path of Sec. B.5. Both realize the same two-stage split-KV decode attention, in which stage 1 fuses the codebook lookup, the per-token scale, and the

Table 6 VQ on the values under two transforms, on RULER NIAH with Qwen3-8B, at fixed group size, codebook size, and rate. Mean accuracy (%) with CIs drawn from 3 evaluation rollouts. SQ corresponds to our deployed method. BPE=2.22 for the SQ method, 2.16 for the VQ method.

Configuration	64K	128K
SQ (deployed)	76.4 $\pm$ 0.4	75.4 $\pm$ 0.1
VQ+ $\mathbf{U}_S$	77.8 $\pm$ 0.2	76.1 $\pm$ 0.5
VQ+ $\mathbf{U}_S$ +Hadamard	76.4 $\pm$ 0.3	76.7 $\pm$ 0.7

attention arithmetic, and stage 2 combines the splits; they share the cache layout and the fp8 codebooks. The *Triton* kernel is the reference implementation, with launch configurations selected offline per batch size and geometry. The *CUDA* kernel is a hand-written stage 1 with a warp-tiled codebook gather and fp32 accumulation; it reduces the instruction count of the gather-and-dequantize loop, which lowers the attention read at batch 128 on Qwen3-8B from 48.9 to 36.4 ms per step (Table 7). The CUDA kernel is the one measured in the main-text throughput comparison (main-text Fig. 4).

Table 7 Per-decoding-step kernel time on Qwen3-8B, one H100, 8192-token contexts. Entries are ms per step with the share of the step in parentheses (%). The BF16 pool admits at most 38 concurrent requests at this length.

Method	B	GEMM	Attn	Quant	Other	Total
BF16	1	5.8 (57.1)	3.8 (37.6)	–	0.5 (5.2)	10.1
	8	5.8 (54.0)	4.4 (40.8)	–	0.6 (5.2)	10.8
	16	5.9 (37.9)	9.0 (58.3)	–	0.6 (3.8)	15.4
	32	5.9 (24.5)	17.5 (73.1)	–	0.6 (2.4)	24.0
	64		<i>capacity limited</i>			
	128		<i>capacity limited</i>			
OSCAR	1	6.0 (72.7)	1.3 (16.4)	0.0 (0.5)	0.9 (10.4)	8.2
	8	6.1 (62.3)	2.6 (26.9)	0.0 (0.5)	1.0 (10.3)	9.8
	16	6.2 (52.8)	4.5 (38.2)	0.1 (0.5)	1.0 (8.5)	11.7
	32	6.2 (39.9)	8.3 (53.2)	0.1 (0.4)	1.0 (6.5)	15.5
	64	6.2 (27.1)	15.7 (68.1)	0.1 (0.4)	1.0 (4.4)	23.1
	128	6.4 (16.2)	31.6 (80.5)	0.1 (0.4)	1.1 (2.9)	39.3
NOVA-KV (Triton)	1	5.8 (67.0)	1.7 (19.5)	0.3 (3.4)	0.9 (10.2)	8.7
	8	5.9 (50.5)	4.1 (35.3)	0.5 (4.5)	1.1 (9.7)	11.7
	16	6.0 (40.6)	6.8 (45.8)	0.8 (5.8)	1.2 (7.9)	14.8
	32	6.0 (28.1)	12.6 (59.5)	1.4 (6.8)	1.2 (5.6)	21.2
	64	6.1 (17.4)	24.7 (70.7)	2.8 (8.0)	1.4 (4.0)	35.0
	128	6.1 (9.9)	48.9 (78.7)	5.3 (8.6)	1.7 (2.8)	62.1
NOVA-KV (CUDA)	1	5.9 (64.8)	2.0 (21.7)	0.3 (3.4)	0.9 (10.0)	9.0
	8	5.9 (55.6)	3.0 (28.1)	0.6 (5.2)	1.2 (11.1)	10.7
	16	6.0 (45.2)	5.2 (39.3)	0.9 (6.6)	1.2 (8.9)	13.3
	32	6.0 (32.8)	9.5 (52.1)	1.5 (8.2)	1.3 (6.9)	18.3
	64	6.1 (21.1)	18.6 (64.4)	2.8 (9.7)	1.4 (4.9)	28.8
	128	6.1 (12.4)	36.4 (73.3)	5.3 (10.8)	1.8 (3.6)	49.7

Per-step kernel profiling We adopt the per-step profiling protocol of Zhou et al. (2026): one server per arm on a single H100, 8192-token contexts, batches 1 to 128, and GPU kernel time per decoding step split by kernel name into *GEMM* (projections, MLP, and MoE experts), *Attn* (the fused attention read, including dequantization and the split combine), *Quant* (the write path: codebook encoder, per-step query preparation, OSCAR pack), and *Other* (norms, activations, rotary, sampling, BF16-band writes). Profiling starts in steady state, after every warmed request is decoding, and stops after exactly 64 forward steps; per-step figures divide device time by the step count recovered from the trace. Tables 7 and 8 report the breakdown. The kernel sum matches the wall clock per step within 1% in every cell, and unassigned kernels are below 0.2%. New tokens enter the BF16 recent band and are quantized in blocks of eight steps in every quantized arm, so the scalar arm’s Quant column is smaller than a per-step fused quantizer would show. The BF16 and OSCAR columns reproduce the profile reported by Zhou et al. (2026) within about 2% at every batch size, which puts

Table 8 Per-decoding-step kernel time on GPT-OSS-20B, same protocol as Table 7.

Method	B	GEMM	Attn	Quant	Other	Total
BF16	1	2.6 (65.5)	0.7 (17.8)	–	0.7 (16.7)	4.0
	8	4.5 (73.2)	0.9 (14.8)	–	0.7 (12.0)	6.2
	16	4.9 (70.1)	1.3 (19.1)	–	0.8 (10.8)	7.1
	32	5.8 (63.7)	2.5 (27.4)	–	0.8 (8.9)	9.1
	64	6.5 (53.8)	4.8 (39.2)	–	0.8 (7.0)	12.1
	128	<i>capacity limited</i>				
OSCAR	1	2.8 (66.0)	0.5 (12.4)	0.0 (0.9)	0.9 (20.8)	4.2
	8	4.2 (68.5)	0.9 (14.8)	0.0 (0.7)	1.0 (15.9)	6.1
	16	4.6 (67.4)	1.2 (17.6)	0.0 (0.6)	1.0 (14.4)	6.9
	32	5.3 (60.5)	2.4 (27.1)	0.0 (0.6)	1.0 (11.8)	8.7
	64	6.2 (52.7)	4.4 (37.6)	0.1 (0.6)	1.1 (9.1)	11.8
	128	7.6 (43.5)	8.6 (49.2)	0.1 (0.4)	1.2 (6.9)	17.5
NOVA-KV (Triton)	1	2.7 (63.4)	0.5 (12.7)	0.1 (2.4)	0.9 (21.5)	4.3
	8	3.3 (56.8)	1.3 (22.5)	0.2 (2.6)	1.1 (18.1)	5.9
	16	3.6 (54.0)	1.7 (25.9)	0.2 (3.4)	1.1 (16.7)	6.6
	32	4.2 (48.6)	3.0 (34.8)	0.3 (3.6)	1.1 (13.0)	8.7
	64	5.2 (40.9)	5.8 (45.2)	0.6 (4.4)	1.2 (9.5)	12.8
	128	6.6 (33.7)	10.5 (53.7)	1.0 (5.1)	1.5 (7.5)	19.6
NOVA-KV (CUDA)	1	2.7 (62.2)	0.6 (14.3)	0.1 (2.4)	0.9 (21.0)	4.4
	8	3.3 (59.8)	1.0 (18.1)	0.2 (2.8)	1.1 (19.4)	5.6
	16	3.5 (54.0)	1.7 (25.6)	0.2 (3.3)	1.1 (17.1)	6.5
	32	4.2 (48.8)	2.9 (34.1)	0.3 (3.8)	1.2 (13.4)	8.6
	64	5.4 (42.1)	5.6 (44.0)	0.5 (4.2)	1.2 (9.7)	12.7
	128	6.5 (32.6)	11.0 (55.2)	1.0 (4.8)	1.5 (7.3)	20.0

the NOVA-KV columns on a verified scale.

Qwen3-8B (Table 7). GEMM is flat at about 6 ms for every arm: parameter load is unaffected by quantization, so the step is a fixed GEMM cost plus an attention read that grows with batch (the regime of main-text Fig. 1). From batch 16 on, every 2-bit arm is faster wherever BF16 runs (at batch 8 the Triton arm’s write-path cost still outweighs the small read saving), and only the 2-bit arms serve batches 64 and 128. At batch 128, NOVA-KV (CUDA) pays $1.15\times$ OSCAR’s attention read (36.4 against 31.6 ms) plus 5.3 ms of write-path cost, for a $1.26\times$ step (49.7 against 39.3 ms); the write-path term grows from 0.3 ms at batch 1 and is the larger contribution beyond batch 64. This is the kernel-level form of the main-text statement that decoding is on par with the scalar baseline: the difference is confined to the read and the write path.

GPT-OSS-20B (Table 8). At this context no 2-bit arm beats BF16 per step: only 12 of 24 layers are quantized, the global-layer cache at 8192 tokens is small, and at $d=64$ the dequantizing read is bound by instruction issue rather than bandwidth, so the write-path cost is not repaid. The MoE GEMMs dominate the step and vary across arms by more than the totals differ (routing depends on the token streams). What quantization buys here is the batch itself: the 2-bit arms serve batch 128, which the BF16 pool cannot hold. The main-text gains for this model come from longer contexts and larger admitted batches, not from per-step speed at 8192 tokens.

B.6 Sampling and statistical testing

Generation is stochastic. For each prompt, we draw five samples on the generation benchmarks and three samples on RULER NIAH. The compression parameters (transforms, groupings, codebooks, the k -means initialization, and the random comparison partition of Sec. 4) are fitted once, with fixed seeds, and held fixed across these samples; the repetitions therefore characterize generation and evaluation variability, not variability from refitting the compression method. Confidence intervals are computed by bootstrap over samples (10000 resamples).

McNemar’s test compares two methods evaluated on the same items when each item’s outcome is binary

(correct/incorrect). For each item, the pair of outcomes falls into one of four cells: both methods correct, both incorrect, only method A correct, or only method B correct. Items where the two methods agree carry no information about their difference, so the test uses only the discordant items: with n_{01} items that only A solves and n_{10} that only B solves, it tests the null hypothesis that a discordant item is equally likely to fall either way, i.e., that the two methods have the same per-item error rate. We use the exact binomial form: under the null, $n_{01} \sim \text{Binomial}(n_{01}+n_{10}, 1/2)$.

We form binary outcomes by majority voting over the samples drawn for a given item (5 on the generation benchmarks, 3 on RULER NIAH), and run the test over the items that both compared methods evaluate on. For the generation benchmarks the five task sets are combined into a single test of 992 paired items per model; on RULER NIAH the test is run at 128K on Qwen3-8B over 200 paired items. Tests are two-sided and reported without any further correction.

B.7 Datasets and licenses

We evaluate on six public benchmarks. *RULER NIAH* (Hsieh et al., 2024) is a synthetic needle-in-a-haystack retrieval suite (Apache-2.0). *GPQA-Diamond* (Rein et al., 2023) is graduate-level multiple-choice science (CC BY 4.0). *HumanEval* (Chen et al., 2021) is Python code synthesis graded by execution (MIT). *LiveCodeBench v6* (Jain et al., 2025) is contamination-controlled competitive programming (benchmark released under MIT; problem statements originate from competition platforms). *AIME25* consists of the 2025 AIME competition problems, copyright © Mathematical Association of America, used here for research evaluation (Mathematical Association of America, 2025). *MATH500* (Lightman et al., 2024; Hendrycks et al., 2021) is a 500-problem subset of the MATH dataset (MIT). The models are Llama-3.1-8B (Llama 3.1 Community License) (Grattafiori et al., 2024), Qwen3-8B / Qwen3-4B-Thinking-2507 (Apache-2.0) (Yang et al., 2025a), and GPT-OSS-20B (Apache-2.0) (Agarwal et al., 2025). All datasets and models are used for research evaluation consistent with their licenses.

B.8 Throughput protocol and tuning

The decode-throughput comparison (main-text Fig. 4) runs one server per arm on a single H100. Prompts are distinct random token streams with no shared prefix. Before the measured pass, a warm-up pass prefills every request’s own prompt into the cache; the measured pass then reuses these prefixes, so all requests enter decoding together and throughput is computed over the decode window alone. Every arm runs its best configuration: OSCAR takes the best of eight decode-kernel tile configurations swept per cell, NOVA-KV (the CUDA kernel, Sec. B.5) takes the best of its compiled variants per cell, and BF16 runs the engine’s dense read with an uncapped memory-fraction pool. All arms share the same chunk size, split-KV budget, and fp32 accumulation in the attention read.

B.9 Configuration summary

Tables 9 and 10 collect the infrastructure, serving, method, and per-benchmark settings described above.

Table 9 Infrastructure, serving, and hyperparameters.

Setting	Value
<i>Software & hardware</i>	
SGLang	research fork based on v0.5.10
GPUs (accuracy)	12×A100 (40GB)
Triton / PyTorch / CUDA (A100)	3.5.1 / 2.9.1 / 12.8
GPUs (throughput, profiling)	1×H100 (80GB)
GPUs (decode-time breakdown)	8×H100 (80GB)
Triton / PyTorch / CUDA (H100)	3.6.0 / 2.11.0 / 13.0
<i>Serving</i>	
Decode attention backend	Triton
Chunked-prefill size	4096 tokens (8192 on GPT-OSS)
Sink / recent band $n_{\text{sink}}, n_{\text{rec}}$	64 / 256 (BF16)
KV cache dtype	INT2, quant group 128 (GPT-OSS: per-row scale, $d=64$)
<i>Transform & quantizer</i>	
Head dim d / group g	128 / 4
Codebook size / rate	256 / 2 bits per coord
Groups L / grouping	32 / volume-equalizing
K / V quantizer	VQ / per-token affine INT2
k -means init / iters	random-distinct / 25
Train. vectors / (layer, head)	131072 (8×16384)
Transform / codebook storage	float64→bf16 / fp16→fp8
NOVA-KV regularizer	eig. floor 10^{-30} (no ridge)
<i>Calibration & decoding</i>	
Calibration	198 GPQA-Diamond, prefill-only
Statistics dtype	float64
<i>Sampling (SGLang serving; each model's released defaults)</i>	
Qwen3-8B, Qwen3-4B-Think.	$T=0.6$, top- p 0.95, top- k 20
Llama-3.1-8B-Instruct	$T=0.6$, top- p 0.9, top- k disabled
GPT-OSS	$T=1.0$, top- p 1.0, top- k disabled
Samples per prompt	5 (3 for NIAH)
Confidence intervals	paired bootstrap 95% (10000 resamples)

Table 10 Per-benchmark evaluation settings. Samples = samples per prompt; Gen = max generated tokens.

Dataset	# items	Smp.	Gen	Metric
RULER NIAH	800 / 200	3	128	substring
GPQA-Diamond	198	5	32768	letter (A–D)
HumanEval	164	5	32768	pass@1 exec.
LiveCodeBench v6	100	5	32768	pass@1 exec.
AIME25	30	5	32768	math-verify
MATH500	500	5	32768	math-verify

C Calibration ablation

We vary the calibration set and refit the transform, grouping, and codebooks (Table 11). Reducing from 198 to 32 GPQA-Diamond prompts (40K to 8.9K tokens) changes RULER NIAH by 1.1 points at 64K and 1.1 at 128K, both upward. Calibrating on MMLU instead, a disjoint domain with 116K tokens, gives 74.4 and 75.4, within 2.0 and 0.0 points of the default. GPQA accuracy under MMLU calibration is 59.3, against 57.2 for BF16 (Table 3).

Table 12 extends this comparison to the five reasoning and coding benchmarks. The effect of the calibration domain is small and model-dependent. Relative to the deployed GPQA-Diamond calibration, MMLU raises the Qwen3-8B mean by 2.0 points but lowers the Qwen3-4B-Thinking mean by 0.7. Both stay within two points of the deployed calibration and within 2.3 points of BF16. For consistency with (Zhou et al., 2026), we use GPQA-Diamond in our main evaluations.

Table 11 Calibration data ablation, Qwen3-8B. We vary the calibration set and token count; GPQA accuracy does not benefit from calibrating on GPQA. Mean accuracy (%) $\pm$ 95% CI over 3 rollouts.

Calibration	Tokens	GPQA	64K	128K
GPQA-198	40000	55.1 $\pm$ 1.5	76.4 $\pm$ 0.4	75.4 $\pm$ 0.1
GPQA-32	8945	56.2 $\pm$ 1.4	77.5 $\pm$ 0.6	76.5 $\pm$ 0.1
MMLU	116340	59.3 $\pm$ 0.4	74.4 $\pm$ 0.3	75.4 $\pm$ 0.6

Table 12 Calibration domain ablation on NOVA-KV, for Qwen3-4B-Thinking-2507 and Qwen3-8B. Compressed rows are NOVA-KV at 2.22 BPE; *GPQA-198* is the deployed calibration, and *MMLU* a disjoint domain (116K tokens). Entries are mean $\pm$ 95% CI with 5 samples per prompt; “Drop” is the gap in the Mean column to the BF16 reference.

Model	Calibration	BPE	GPQA	HumanE	LCB v6	AIME25	MATH500	Mean	Drop
Qwen3-4B -Thinking-2507	BF16	16.00	64.9 $\pm$ 0.8	84.8 $\pm$ 0.5	59.2 $\pm$ 0.9	70.0 $\pm$ 1.5	97.5 $\pm$ 0.2	75.3	–
	GPQA-198	2.22	62.9 $\pm$ 1.3	84.3 $\pm$ 0.4	56.6 $\pm$ 0.6	67.3 $\pm$ 1.2	97.4 $\pm$ 0.2	73.7	–1.6
	MMLU	2.22	64.7 $\pm$ 1.2	83.7 $\pm$ 1.6	53.0 $\pm$ 3.8	66.7 $\pm$ 8.3	97.0 $\pm$ 0.5	73.0	–2.3
Qwen3-8B	BF16	16.00	57.2 $\pm$ 0.9	91.7 $\pm$ 0.5	57.6 $\pm$ 0.7	70.0 $\pm$ 1.7	97.0 $\pm$ 0.1	74.7	–
	GPQA-198	2.22	55.1 $\pm$ 1.5	92.1 $\pm$ 0.2	55.6 $\pm$ 0.6	64.6 $\pm$ 2.5	96.9 $\pm$ 0.2	72.9	–1.8
	MMLU	2.22	57.5 $\pm$ 3.3	91.5 $\pm$ 1.4	60.0 $\pm$ 2.3	68.7 $\pm$ 6.3	96.7 $\pm$ 0.5	74.9	+0.2

D Partition statistics

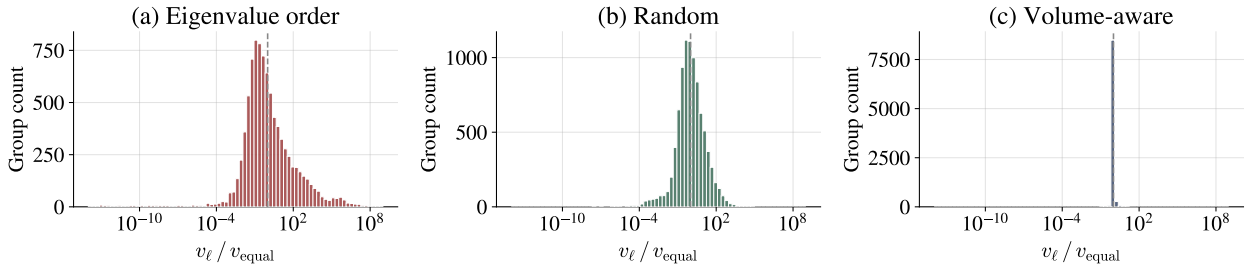


Figure 7 Distribution of group volumes under the three partitions, averaged over all (layer, KV head) pairs of Qwen3-8B at $g = 4$. Each count is one group; volumes are normalized by v_{equal} , the common value they would take under a perfectly volume-equalizing partition (dashed line). The horizontal axis is logarithmic.

The fixed-rate case of Theorem 2 bounds the distortion of a partition π by $C_g 2^{-2b} \sum_{\ell} v_{\ell}(\pi)^{1/g} \geq D^*(b)$, with equality iff all group volumes coincide. In theory, the gap is a property of the volume distribution (cf. Sec. H). Fig. 7 shows that distribution for the three partitions compared in Sec. 5. Volumes are normalized by:

$$v_{\text{equal}} = \left(\prod_{i=1}^d \sigma_i^2 \right)^{g/d}, \quad (9)$$

the group volume under an exactly equalizing partition; hence, a partition that equalizes exactly places all mass on 1.

The three panels differ by orders of magnitude in spread. Grouping consecutive entries in eigenvalue order (a) produces volumes spanning roughly 20 decades, since the leading group collects the largest σ_i^2 and the trailing group the smallest. A random partition (b) is narrower, as each group draws entries from across the spectrum, but the spread remains close to 3 decades. The volume-aware partition (c) concentrates almost all groups with a small tail attributable to the greedy dealing rule.

The ordering of the three spreads matches the ordering of the corresponding accuracies reported in Sec. 5, which is consistent with Theorem 2. We note that the theorem is derived under the high-resolution model with independent Gaussian coefficients, so the agreement is evidence that the predicted ordering survives outside the regime of the derivation rather than a verification of the bound (cf. Sec. H).

E Analytical tools

E.1 Companding for non-difference distortions

This subsection states the high-resolution theory for distortion measures $\rho(\cdot, \cdot)$ other than the MSE, following the companding framework of Linder et al. (1999), which covers locally quadratic measures whose weight may be input-dependent. The criterion in (5), that is,

$$\|\mathbf{Q}\mathbf{K}^\top - \mathbf{Q}\hat{\mathbf{K}}^\top\|_F^2 = \sum_{j=1}^M (\mathbf{k}_j - \hat{\mathbf{k}}_j) \mathbf{M}_q (\mathbf{k}_j - \hat{\mathbf{k}}_j)^\top, \quad (10)$$

is a special case of this theory, as we will show next.

Locally quadratic distortions. Suppose $\rho(\cdot, \cdot)$ is three times continuously differentiable in its second argument, vanishes exactly when its second argument equals the first argument, and has positive definite second derivative at that point. Define the *sensitivity matrix*

$$[\mathbf{M}(\mathbf{x})]_{ij} = \frac{1}{2} \frac{\partial^2 \rho(\mathbf{x}, \hat{\mathbf{x}})}{\partial \hat{x}_i \partial \hat{x}_j} \Big|_{\hat{\mathbf{x}}=\mathbf{x}}. \quad (11)$$

Because the gradient vanishes at $\hat{\mathbf{x}} = \mathbf{x}$, a second-order expansion gives

$$\rho(\mathbf{x}, \hat{\mathbf{x}}) = (\mathbf{x} - \hat{\mathbf{x}}) \mathbf{M}(\mathbf{x}) (\mathbf{x} - \hat{\mathbf{x}})^\top + O(\|\mathbf{x} - \hat{\mathbf{x}}\|^3), \quad (12)$$

so at high resolution the distortion is a quadratic form with a possibly input-dependent weight. An input-weighted quadratic measure

$$\rho(\mathbf{x}, \hat{\mathbf{x}}) = \|(\mathbf{x} - \hat{\mathbf{x}}) \mathbf{W}(\mathbf{x})\|_2^2, \quad (13)$$

is of this type, with $\mathbf{M}(\mathbf{x}) = \mathbf{W}(\mathbf{x}) \mathbf{W}(\mathbf{x})^\top$. In our criterion (5), the weight does not depend on the input, $\mathbf{M}(\mathbf{x}) \equiv \mathbf{M}_q$, and the expansion (12) is exact.

Companding quantizers. A compander applies an invertible map h to the source, quantizes with a lattice quantizer $\mathcal{Q}_\mathcal{L}$, and inverts:

$$\mathbf{x} \longrightarrow h(\mathbf{x}) \longrightarrow \mathcal{Q}_\mathcal{L}(h(\mathbf{x})) \longrightarrow h^{-1} \longrightarrow \hat{\mathbf{x}}. \quad (14)$$

Write $\mathbf{J}(\mathbf{x})$ for the Jacobian of h , so that $dh = d\mathbf{x} \mathbf{J}(\mathbf{x})$. The compander is a structured vector quantizer: all adaptation to the source and to the distortion is carried by h , while the cells in the companded domain are congruent, i.e., translates of the basic lattice cell, identical in shape and volume. Mapped back through h^{-1} , these identical cells become the non-uniform cells of the effective quantizer in the source domain. A linear $h(\mathbf{x}) = \mathbf{x} \mathbf{R}$, with $\mathbf{J} \equiv \mathbf{R}$ a typical transform coder. In this case, $dh = d\mathbf{x} \mathbf{R}$.

Asymptotic rate. Let Z_g denote the normalized second moment of the basic cell of the lattice. Linder et al. (1999) show that, for a source with finite differential entropy $H(\mathbf{x})$, the rate of the compander, defined as the entropy $\mathcal{H}(D)$ of the quantizer indices when the lattice is scaled to operate at distortion D , satisfies

$$\lim_{D \rightarrow 0} \left(\mathcal{H}(D) + g/2 \log_2 D \right) = H(\mathbf{x}) + \mathbb{E}[\log_2 |\det \mathbf{J}(\mathbf{x})|] + g/2 \log_2 \left(g Z_g \mathbb{E}[\text{tr } \mathbf{\Gamma}(\mathbf{x})] \right), \quad (15)$$

where

$$\mathbf{\Gamma}(\mathbf{x}) = \mathbf{J}(\mathbf{x})^{-1} \mathbf{M}(\mathbf{x}) \mathbf{J}(\mathbf{x})^{-\top}, \quad (16)$$

measures the sensitivity as seen in the companded domain.

The optimal compressor. Minimizing (15) over h uses two inequalities. Since $\mathbf{\Gamma}(\mathbf{x}) \succ 0$, the arithmetic–geometric mean inequality gives $\text{tr } \mathbf{\Gamma} \geq g (\det \mathbf{\Gamma})^{1/g}$, with equality iff the eigenvalues of $\mathbf{\Gamma}$ are all equal; Jensen’s inequality then moves the expectation inside the logarithm. Together (Linder et al., 1999, Thm. 2):

$$\lim_{D \rightarrow 0} \left(\mathcal{H}(D) + g/2 \log_2 D \right) \geq H(\mathbf{x}) + g/2 \log_2 (g Z_g) + 1/2 \mathbb{E}[\log_2 \det \mathbf{M}(\mathbf{x})], \quad (17)$$

with equality if and only if

$$\mathbf{J}(\mathbf{x}) \mathbf{J}(\mathbf{x})^\top = c \mathbf{M}(\mathbf{x}) \quad \text{a.e., for some } c > 0. \quad (18)$$

Three consequences:

- (C1) *The optimal compressor does not depend on the source distribution*, only on the distortion measure. This is the analogue, for locally quadratic distortions, of the fact that the optimal entropy-coded quantizer for MSE is uniform regardless of the source.
- (C2) *Condition (18) fixes h only up to an orthogonal factor on the right*, since replacing $\mathbf{J}$ by $\mathbf{J}\mathbf{O}$ with $\mathbf{O}$ orthogonal leaves $\mathbf{J}\mathbf{J}^\top$ unchanged. The condition therefore constrains the stretch but not the rotation.
- (C3) *An orthogonal h satisfies (18) only when $\mathbf{M} \propto \mathbf{I}$* , since $\mathbf{J}\mathbf{J}^\top = \mathbf{I}$ then forces $\mathbf{M} = c^{-1}\mathbf{I}$.

For the key transform. Take $\mathbf{M}(\mathbf{x}) \equiv \mathbf{M}_q$, constant. The compressor is then linear, $h(\mathbf{x}) = \mathbf{x}\mathbf{R}$, and (18) reads $\mathbf{R}\mathbf{R}^\top = c\mathbf{M}_q$. By (C3), orthogonal transforms are suboptimal for the attention-aware criterion unless the queries are isotropic; by (C2), Theorem 1 must resolve the remaining orthogonal factor, which we do by minimizing the rank- p reconstruction error.

E.2 Softmax Perturbation Bound

We prove that the logit error controls the attention weights (beginning of Sec. 3).

Proposition S1. *Let $\mathbf{Z}, \hat{\mathbf{Z}} \in \mathbb{R}^{T \times T}$, $\mathbf{S} = \text{softmax}_{\text{row}}(\mathbf{Z}/\sqrt{d})$, and $\hat{\mathbf{S}} = \text{softmax}_{\text{row}}(\hat{\mathbf{Z}}/\sqrt{d})$. Then, $\|\mathbf{S} - \hat{\mathbf{S}}\|_F \leq 1/(2\sqrt{d}) \|\mathbf{Z} - \hat{\mathbf{Z}}\|_F$.*

Proof. Consider one row. The Jacobian of softmax at $\mathbf{z}$ is $\mathbf{J}(\mathbf{z}) = \text{diag}(\mathbf{s}) - \mathbf{s}^\top \mathbf{s}$, $\mathbf{s} = \text{softmax}(\mathbf{z})$, symmetric positive semidefinite with $\|\mathbf{J}(\mathbf{z})\|_2 \leq 1/2$ (Gao and Pavel, 2017). Along the segment $\mathbf{z}(t) = (1-t)\hat{\mathbf{z}} + t\mathbf{z}$,

$$\|\text{softmax}(\mathbf{z}) - \text{softmax}(\hat{\mathbf{z}})\|_2 \leq \sup_{t \in [0,1]} \|\mathbf{J}(\mathbf{z}(t))\|_2 \|\mathbf{z} - \hat{\mathbf{z}}\|_2 \leq 1/2 \|\mathbf{z} - \hat{\mathbf{z}}\|_2. \quad (19)$$

Applying this to each row of $\mathbf{Z}/\sqrt{d}$, $\hat{\mathbf{Z}}/\sqrt{d}$ and summing squares over rows gives the claim. $\square$

With $\mathbf{Z} = \mathbf{Q}\mathbf{K}^\top$ and $\hat{\mathbf{Z}} = \mathbf{Q}\hat{\mathbf{K}}^\top$, the attention-weight error is bounded by $1/(2\sqrt{d}) \|\mathbf{Q}\mathbf{K}^\top - \mathbf{Q}\hat{\mathbf{K}}^\top\|_F$.

E.3 Decoupling of the Output Error

We justify the decomposition of Sec. 3. Let $\hat{\mathbf{S}} = \text{softmax}_{\text{row}}(\mathbf{Q}\hat{\mathbf{K}}^\top/\sqrt{d})$ be the scores computed from reconstructed keys and $\hat{\mathbf{O}} = \hat{\mathbf{S}}\hat{\mathbf{V}}$ the output under joint key and value quantization.

Proposition S2.

$$\|\mathbf{O} - \hat{\mathbf{O}}\|_F \leq \|\mathbf{V}\|_2/(2\sqrt{d}) \|\mathbf{Q}\mathbf{K}^\top - \mathbf{Q}\hat{\mathbf{K}}^\top\|_F + \|\mathbf{S}(\mathbf{V} - \hat{\mathbf{V}})\|_F + \|(\mathbf{S} - \hat{\mathbf{S}})(\mathbf{V} - \hat{\mathbf{V}})\|_F. \quad (20)$$

Proof. Adding and subtracting $\hat{\mathbf{S}}\hat{\mathbf{V}}$ gives the exact identity

$$\mathbf{O} - \hat{\mathbf{O}} = (\mathbf{S} - \hat{\mathbf{S}})\mathbf{V} + \mathbf{S}(\mathbf{V} - \hat{\mathbf{V}}) - (\mathbf{S} - \hat{\mathbf{S}})(\mathbf{V} - \hat{\mathbf{V}}). \quad (21)$$

The triangle inequality, $\|(\mathbf{S} - \hat{\mathbf{S}})\mathbf{V}\|_F \leq \|\mathbf{S} - \hat{\mathbf{S}}\|_F \|\mathbf{V}\|_2$, and Proposition S1 give the claim. $\square$

The first term is controlled by the key objective (5); the second is the value objective of Corollary 1 (Sec. F.4); the third is a product of the two quantization errors, hence second order. We do not claim joint optimality of the two-step design for the combined objective. Consistently with this decomposition, the value objective below treats the scores as computed from full-precision keys; the discrepancy from using $\hat{\mathbf{S}}$ instead is absorbed by the second-order term.

F Optimal transforms

F.1 Lemma S1: Softmax Offset Invariance

We prove in this section that adding a constant to all keys leaves the attention scores unaltered.

Lemma S1. For any $\mathbf{c} \in \mathbb{R}^{1 \times d}$ and any query $\mathbf{q}$, $\text{softmax}_j(\mathbf{q}(\mathbf{k}_j - \mathbf{c})^\top) = \text{softmax}_j(\mathbf{q}\mathbf{k}_j^\top)$.

Proof. Replacing each key by $\mathbf{k}_j - \mathbf{c}$ shifts every logit of the query $\mathbf{q}$ by the same amount $\mathbf{q}\mathbf{c}^\top$, which cancels:

$$\frac{\exp(\mathbf{q}(\mathbf{k}_j - \mathbf{c})^\top)}{\sum_{j'} \exp(\mathbf{q}(\mathbf{k}_{j'} - \mathbf{c})^\top)} = \frac{\exp(\mathbf{q}\mathbf{k}_j^\top)}{\sum_{j'} \exp(\mathbf{q}\mathbf{k}_{j'}^\top)}. \quad (22)$$

$\square$

By Lemma S1, replacing every key $\mathbf{k}_j$ by $\mathbf{k}_j - \mathbf{c}$ leaves the attention unchanged, so we may quantize the shifted keys instead and minimize

$$\min_{\mathbf{c}, \mathbf{D}, \{\mathbf{r}_j\}} \sum_{j=1}^M \|\mathbf{k}_j - \mathbf{c} - \mathbf{r}_j \mathbf{D}\|_{\mathbf{M}_q}^2, \quad (23)$$

which coincides with the original objective at $\mathbf{c} = \mathbf{0}$; optimizing over $\mathbf{c}$ therefore never increases the error, and we choose $\mathbf{c}$ to make the keys cheaper to compress.

F.2 Theorem 1: Key Transform

We find the transform that minimizes the low-rank reconstruction $\mathbf{M}_q$ -MSE given training examples. We write $\mathbf{R}^\dagger = (\mathbf{R}^\top \mathbf{M}_q^{-1} \mathbf{R})^{-1} \mathbf{R}^\top \mathbf{M}_q^{-1}$ for the $\mathbf{M}_q$ -weighted pseudoinverse, so that $\mathbf{R}\mathbf{R}^\dagger$ is the projector onto the column space of $\mathbf{R}$ that is orthogonal in the $\mathbf{M}_q$ inner product.

Theorem 1 (restated). Let $\tilde{\mathbf{k}}_j = \mathbf{k}_j - \bar{\mathbf{k}}$, $\tilde{\mathbf{S}}_k = \sum_j \tilde{\mathbf{k}}_j^\top \tilde{\mathbf{k}}_j$, and let

$$\mathbf{R}_{K,p}^* = \arg \min_{\mathbf{R} \in \mathbb{R}^{d \times p}} \sum_{j=1}^M \|\tilde{\mathbf{k}}_j - \tilde{\mathbf{k}}_j \mathbf{R} \mathbf{R}^\dagger\|_{\mathbf{M}_q}^2.$$

Then $\mathbf{R}_{K,p}^* = \mathbf{M}_q^{1/2} \mathbf{E}_{1,p}$, where $\mathbf{E} \mathbf{\Lambda} \mathbf{E}^\top$ is the eigendecomposition of $\mathbf{M}_q^{1/2} \tilde{\mathbf{S}}_k \mathbf{M}_q^{1/2}$.

We prove a slightly more general statement. Rather than fixing the reconstruction to the $\mathbf{M}_q$ -orthogonal projection $\tilde{\mathbf{k}}_j \mathbf{R} \mathbf{R}^\top$, we leave the codes, the synthesis matrix, and a shared offset free, and minimize the logit error (5) directly. In particular, we parameterize a rank- p linear reconstruction with codes $\mathbf{r}_j \in \mathbb{R}^{1 \times p}$, a shared synthesis matrix $\mathbf{D} \in \mathbb{R}^{p \times d}$ of full row rank, and a shared offset $\mathbf{c} \in \mathbb{R}^{1 \times d}$:

$$\hat{\mathbf{k}}_j = \mathbf{r}_j \mathbf{D} + \mathbf{c}, \quad (24)$$

and minimize

$$L(\mathbf{D}, \mathbf{c}, \{\mathbf{r}_j\}) = \sum_{i=1}^N \sum_{j=1}^M (\mathbf{q}_i \mathbf{k}_j^\top - \mathbf{q}_i \hat{\mathbf{k}}_j^\top)^2. \quad (25)$$

We show our result in 5 steps.

Step 1: reorder to expose the query second moment. With $\boldsymbol{\delta}_j = \mathbf{k}_j - \mathbf{c} - \mathbf{r}_j \mathbf{D}$,

$$\sum_{i,j} (\mathbf{q}_i \boldsymbol{\delta}_j^\top)^2 = \sum_{j=1}^M \boldsymbol{\delta}_j \left(\sum_{i=1}^N \mathbf{q}_i^\top \mathbf{q}_i \right) \boldsymbol{\delta}_j^\top = \sum_{j=1}^M \|\boldsymbol{\delta}_j\|_{\mathbf{M}_q}^2, \quad (26)$$

with $\mathbf{M}_q = \mathbf{Q}^\top \mathbf{Q}$, which is (5). Invertibility of $\mathbf{M}_q$ requires the calibration queries to span $\mathbb{R}^d$ (in particular $N \geq d$), which holds for typical calibration sets; in practice we enforce positive-definiteness as in Appendix A.3.

Step 2: optimal codes for fixed $\mathbf{D}, \mathbf{c}$. The loss decouples across j . Setting the gradient of $\|\boldsymbol{\delta}_j\|_{\mathbf{M}_q}^2$ in $\mathbf{r}_j$ to zero,

$$\mathbf{r}_j \mathbf{D} \mathbf{M}_q \mathbf{D}^\top = (\mathbf{k}_j - \mathbf{c}) \mathbf{M}_q \mathbf{D}^\top \implies \mathbf{r}_j^* = (\mathbf{k}_j - \mathbf{c}) \mathbf{M}_q \mathbf{D}^\top (\mathbf{D} \mathbf{M}_q \mathbf{D}^\top)^{-1}, \quad (27)$$

a generalized least-squares projection onto the row space of $\mathbf{D}$, measured in the $\mathbf{M}_q$ inner product.

Step 3: optimal offset. For fixed $\mathbf{D}$ and codes, $\nabla_{\mathbf{c}} L = -2 \sum_j \boldsymbol{\delta}_j \mathbf{M}_q = \mathbf{0}$. Since $\mathbf{M}_q \succ 0$, the condition reduces to $\sum_j \boldsymbol{\delta}_j = \mathbf{0}$, i.e., $\mathbf{c}^* = \bar{\mathbf{k}} - \bar{\mathbf{r}} \mathbf{D}$ with $\bar{\mathbf{r}} = 1/M \sum_j \mathbf{r}_j$. Any choice satisfying this equation is optimal; we pick the one that centers the codes, $\bar{\mathbf{r}} = \mathbf{0}$, so that no rate is spent encoding the code mean $\mathbf{c}^* = \bar{\mathbf{k}}$. We write $\tilde{\mathbf{k}}_j = \mathbf{k}_j - \bar{\mathbf{k}}$.

Step 4: optimal synthesis. Substituting (27), $\mathbf{r}_j^* \mathbf{D} = \tilde{\mathbf{k}}_j \mathbf{P}$ with $\mathbf{P} = \mathbf{M}_q \mathbf{D}^\top (\mathbf{D} \mathbf{M}_q \mathbf{D}^\top)^{-1} \mathbf{D}$, the oblique projector onto the row space of $\mathbf{D}$ that is orthogonal in the $\mathbf{M}_q$ inner product ($\mathbf{P}^2 = \mathbf{P}$, and $\mathbf{P} \mathbf{M}_q$ is symmetric). Using these two identities, $(\mathbf{I} - \mathbf{P}) \mathbf{M}_q (\mathbf{I} - \mathbf{P})^\top = \mathbf{M}_q - \mathbf{P} \mathbf{M}_q$, and with $\tilde{\mathbf{S}}_k = \sum_j \tilde{\mathbf{k}}_j^\top \tilde{\mathbf{k}}_j$,

$$L^*(\mathbf{D}) = \text{tr}[\mathbf{M}_q \tilde{\mathbf{S}}_k] - \text{tr}[\mathbf{P} \mathbf{M}_q \tilde{\mathbf{S}}_k]. \quad (28)$$

The first term is constant, so we maximize $T(\mathbf{D}) = \text{tr}[\mathbf{M}_q \mathbf{D}^\top (\mathbf{D} \mathbf{M}_q \mathbf{D}^\top)^{-1} \mathbf{D} \mathbf{M}_q \tilde{\mathbf{S}}_k]$.

Step 5: change of variables and Ky Fan. Let $\mathbf{B} = \mathbf{D} \mathbf{M}_q^{1/2} \in \mathbb{R}^{p \times d}$, so $\mathbf{D} \mathbf{M}_q \mathbf{D}^\top = \mathbf{B} \mathbf{B}^\top$ and $\mathbf{M}_q \mathbf{D}^\top = \mathbf{M}_q^{1/2} \mathbf{B}^\top$. Then $T = \text{tr}[\mathbf{\Pi}_B \mathbf{A}]$, $\mathbf{A} = \mathbf{M}_q^{1/2} \tilde{\mathbf{S}}_k \mathbf{M}_q^{1/2}$, $\mathbf{\Pi}_B = \mathbf{B}^\top (\mathbf{B} \mathbf{B}^\top)^{-1} \mathbf{B}$, where $\mathbf{\Pi}_B$ is a rank- p orthogonal projector and $\mathbf{A} \succeq 0$. Maximizing $\text{tr}[\mathbf{\Pi}_B \mathbf{A}]$ over rank- p orthogonal projectors is the classical PCA problem, solved by the projector onto the top- p eigenvectors of $\mathbf{A}$ with maximum $\sum_{i \leq p} \lambda_i(\mathbf{A})$ (Fan, 1949). With $\mathbf{A} = \mathbf{E} \boldsymbol{\Lambda} \mathbf{E}^\top$ and $\mathbf{E}_{1:p}$ the top- p eigenvectors, take $\mathbf{B}^* = \mathbf{E}_{1:p}^\top$, i.e.,

$$\mathbf{D}^* = \mathbf{E}_{1:p}^\top \mathbf{M}_q^{-1/2} = \mathbf{R}^\top. \quad (29)$$

Then $\mathbf{D}^* \mathbf{M}_q \mathbf{D}^{*\top} = \mathbf{I}_p$, and (27) simplifies to

$$\mathbf{r}_j^* = \tilde{\mathbf{k}}_j \mathbf{M}_q \mathbf{D}^{*\top} = \tilde{\mathbf{k}}_j \mathbf{M}_q^{1/2} \mathbf{E}_{1:p} = \tilde{\mathbf{k}}_j \mathbf{R}, \quad (30)$$

with $\mathbf{R} = \mathbf{M}_q^{1/2} \mathbf{E}_{1:p} = \mathbf{R}_{K,p}^*$. The reconstruction is therefore $\hat{\mathbf{k}}_j = \tilde{\mathbf{k}}_j \mathbf{R} \mathbf{R}^\top + \bar{\mathbf{k}}$, the projected form of the theorem, and the optimal cost is $\sum_{i > p} \lambda_i(\mathbf{A})$. For $p = d$, $\mathbf{E}$ is orthogonal and $\mathbf{R} \mathbf{R}^\top = \mathbf{I}$. ■

Remark S1. If $\mathbf{M}_q \propto \mathbf{I}$, the procedure reduces to ordinary mean-centered PCA on the keys.

Remark S2. For $p = d$, the columns of $\mathbf{R}$ are not orthonormal in general; they satisfy $\mathbf{R}\mathbf{R}^\top = \mathbf{M}_q \neq \mathbf{I}$ whenever $\mathbf{M}_q \not\propto \mathbf{I}$. For $p < d$, $\mathbf{R}\mathbf{R}^\top = \mathbf{M}_q^{1/2} \mathbf{E}_{1:p} \mathbf{E}_{1:p}^\top \mathbf{M}_q^{1/2} \neq \mathbf{M}_q$.

F.3 Proposition 1: Generalized Parseval Relation

Throughout, $\mathbf{R}_K = \mathbf{R}_{K,d}^*$ denotes the full-rank ($p = d$) transform of Theorem 1, which is invertible.

Proposition 1 (restated). Let $\mathbf{r}$ and $\hat{\mathbf{r}}$ be any two vectors, and $\mathbf{k} = \mathbf{r}\mathbf{R}_K^{-1}$ and $\hat{\mathbf{k}} = \hat{\mathbf{r}}\mathbf{R}_K^{-1}$. Then, $\|\mathbf{r} - \hat{\mathbf{r}}\|_2^2 = \|\mathbf{k} - \hat{\mathbf{k}}\|_{\mathbf{M}_q}^2$.

Proof. With $\mathbf{r} - \hat{\mathbf{r}} = (\mathbf{k} - \hat{\mathbf{k}})\mathbf{R}$,

$$\|\mathbf{r} - \hat{\mathbf{r}}\|_2^2 = (\mathbf{k} - \hat{\mathbf{k}}) \mathbf{R} \mathbf{R}^\top (\mathbf{k} - \hat{\mathbf{k}})^\top, \quad (31)$$

and $\mathbf{R}\mathbf{R}^\top = \mathbf{M}_q^{1/2} \mathbf{E} \mathbf{E}^\top \mathbf{M}_q^{1/2} = \mathbf{M}_q$, since $\mathbf{E}$ is orthogonal for $p = d$. $\square$

F.4 Corollary 1: Value Transform

Corollary 1 (restated). Let $\mathbf{o}_i = \mathbf{s}_i \mathbf{V}$ denote the attention outputs on calibration data, and $\mathbf{E} \mathbf{\Lambda} \mathbf{E}^\top$ the eigendecomposition of $\mathbf{M}_o = \mathbf{V}^\top \mathbf{M}_s \mathbf{V}$. The minimizer of $\|\mathbf{S} \mathbf{V} - \widehat{\mathbf{S}} \mathbf{V}\|_F^2$ over low-rank approximations is $\mathbf{R}_V = \mathbf{E}_{1:p}$, with codes $\mathbf{s}_j = \mathbf{v}_j \mathbf{R}_V$ and reconstruction $\hat{\mathbf{v}}_j = \mathbf{s}_j \mathbf{R}_V^\top$.

Let $\alpha_i \in \mathbb{R}^{1 \times M}$ denote the i th row of $\mathbf{S}$, so the attention output for calibration query i is $\mathbf{o}_i = \alpha_i \mathbf{V} \in \mathbb{R}^{1 \times d}$. We parameterize a rank- p linear reconstruction with codes $\mathbf{s}_j \in \mathbb{R}^{1 \times p}$ and synthesis $\mathbf{D}_V \in \mathbb{R}^{p \times d}$ of full row rank, $\hat{\mathbf{v}}_j = \mathbf{s}_j \mathbf{D}_V$, and minimize the output distortion

$$L(\mathbf{D}_V, \{\mathbf{s}_j\}) = \sum_{i=1}^N \|\alpha_i \mathbf{V} - \alpha_i \hat{\mathbf{v}}\|_2^2. \quad (32)$$

Step 1: reorder to expose the score second moment. With $\delta_j = \mathbf{v}_j - \mathbf{s}_j \mathbf{D}_V$ and $\Delta \in \mathbb{R}^{M \times d}$ stacking the δ_j as rows,

$$\sum_i \left\| \sum_j \alpha_{ij} \delta_j \right\|_2^2 = \sum_{j,j'} (\mathbf{M}_s)_{jj'} \delta_j \delta_{j'}^\top = \text{tr}[\Delta^\top \mathbf{M}_s \Delta], \quad (33)$$

with $\mathbf{M}_s = \mathbf{S}^\top \mathbf{S} = \sum_i \alpha_i^\top \alpha_i$. Attention reads values through score-weighted sums (mixing tokens).

Step 2: optimal codes for fixed $\mathbf{D}_V$. Writing $\Delta = \mathbf{V} - \mathbf{S}_c \mathbf{D}_V$ with $\mathbf{S}_c \in \mathbb{R}^{M \times p}$ stacking the codes gives

$$\mathbf{M}_s \mathbf{S}_c \mathbf{D}_V \mathbf{D}_V^\top = \mathbf{M}_s \mathbf{V} \mathbf{D}_V^\top. \quad (34)$$

A solution independent of $\mathbf{M}_s$ is

$$\mathbf{S}_c^* = \mathbf{V} \mathbf{D}_V^\top (\mathbf{D}_V \mathbf{D}_V^\top)^{-1}, \quad (35)$$

the ordinary Euclidean projection of each $\mathbf{v}_j$ onto the row space of $\mathbf{D}_V$: the weight lives on the token axis, orthogonal to the head-space axis along which the projection acts, so the projection geometry is standard.¹

Step 3: optimal synthesis. Substituting, $\mathbf{S}_c^* \mathbf{D}_V = \mathbf{V} \mathbf{P}$ with $\mathbf{P} = \mathbf{D}_V^\top (\mathbf{D}_V \mathbf{D}_V^\top)^{-1} \mathbf{D}_V$, a standard rank- p orthogonal projector. Then

$$\text{tr}[(\mathbf{I} - \mathbf{P}) \mathbf{V}^\top \mathbf{M}_s \mathbf{V} (\mathbf{I} - \mathbf{P})] = \text{tr}[\mathbf{M}_o] - \text{tr}[\mathbf{P} \mathbf{M}_o], \quad (36)$$

with $\mathbf{M}_o = \mathbf{V}^\top \mathbf{M}_s \mathbf{V}$. Maximizing $\text{tr}[\mathbf{P} \mathbf{M}_o]$ over rank- p orthogonal projectors is again the PCA problem of Sec. F.2, Step 5: with $\mathbf{M}_o = \mathbf{E} \mathbf{\Lambda} \mathbf{E}^\top$, the maximizer is $\mathbf{P}^* = \mathbf{E}_{1:p} \mathbf{E}_{1:p}^\top$, i.e., $\mathbf{D}_V^* = \mathbf{E}_{1:p}^\top$ and, by (35), $\mathbf{s}_j^* = \mathbf{v}_j \mathbf{E}_{1:p}$. This is Corollary 1 with $\mathbf{R}_V = \mathbf{E}_{1:p}$; since $\mathbf{E}_{1:p}$ has orthonormal columns, the synthesis is the transpose and no normalization is needed.

¹If $\mathbf{M}_s$ is invertible, (35) is the unique solution. Invertibility is not required for what follows: the optimal basis depends on $\mathbf{M}_s$ only through the $d \times d$ matrix $\mathbf{M}_o = \mathbf{V}^\top \mathbf{M}_s \mathbf{V}$, which has rank $\min(d, N)$ regardless of the cache length M .

Second moment of the attention outputs. Although $\mathbf{M}_o$ was defined through the $M \times M$ matrix $\mathbf{M}_s$, it is the empirical second moment of the attention outputs:

$$\mathbf{V}^\top \left(\sum_i \boldsymbol{\alpha}_i^\top \boldsymbol{\alpha}_i \right) \mathbf{V} = \sum_{i=1}^N (\boldsymbol{\alpha}_i \mathbf{V})^\top (\boldsymbol{\alpha}_i \mathbf{V}) = \sum_{i=1}^N \mathbf{o}_i^\top \mathbf{o}_i. \quad (37)$$

We never form $\mathbf{M}_s$ on the token axis, which grows with context length; we accumulate outer products of attention-output vectors in head space, a $d \times d$ matrix. ■

Remark S3. *No offset appears in Corollary 1, but centering the values is also free: since $\mathbf{S}\mathbf{1} = \mathbf{1}$, a shared value offset satisfies $\mathbf{S}(\mathbf{V} - \mathbf{1}\mathbf{c}_V) = \mathbf{S}\mathbf{V} - \mathbf{1}\mathbf{c}_V$, so it can be subtracted before quantization and added back to the output exactly.*

F.5 Query Statistics under Grouped-Query Attention

In grouped-query attention (GQA) (Ainslie et al., 2023), one KV head is shared by H query heads with queries $\mathbf{Q}^{(1)}, \dots, \mathbf{Q}^{(H)}$. The logit error for the shared keys is

$$\sum_{h=1}^H \|\mathbf{Q}^{(h)} \mathbf{K}^\top - \mathbf{Q}^{(h)} \hat{\mathbf{K}}^\top\|_F^2 = \sum_{j=1}^M (\mathbf{k}_j - \hat{\mathbf{k}}_j) \left(\sum_{h=1}^H \mathbf{M}_q^{(h)} \right) (\mathbf{k}_j - \hat{\mathbf{k}}_j)^\top, \quad (38)$$

with $\mathbf{M}_q^{(h)} = \mathbf{Q}^{(h)\top} \mathbf{Q}^{(h)}$. This has the same form as (5) with $\mathbf{M}_q = \sum_h \mathbf{M}_q^{(h)}$, so we accumulate the query second moment over the heads sharing each KV head.

G Quantizer design: grouping and rate allocation

G.1 Theorem 2: Grouping

Theorem 2 (restated). *Under the model (7), for any partition π the allocation minimizing $D(\pi, \{b_\ell\})$ subject to $\sum_\ell b_\ell = Lb$ is*

$$b_\ell^*(\pi) = b + \frac{1}{2g} \log_2 \left(v_\ell(\pi) / \prod_m v_m(\pi)^{1/L} \right),$$

which spends more bits on groups of larger volume. The resulting optimal distortion, $D^(b) = C_g L 2^{-2b} (\prod_{i=1}^d \sigma_i^2)^{1/d}$, is the same for every π , assuming that rates can be the arbitrary real values from (8).*

We prove this statement together with the fixed-rate claim that follows Theorem 2 in the main text: under $b_\ell \equiv b$, the distortion of a partition π is $C_g 2^{-2b} \sum_\ell v_\ell(\pi)^{1/g} \geq D^*(b)$, with equality iff $v_1(\pi) = \dots = v_L(\pi)$.

By the model (7) the total distortion is $D(\pi, \{b_\ell\}) = C_g \sum_\ell 2^{-2b_\ell} v_\ell(\pi)^{1/g}$; the constant $C_g > 0$ is common to all groups, so it leaves the minimizing π and $\{b_\ell\}$ unchanged and only rescales the optimal value. We therefore solve

$$\min_{\pi} \min_{\{b_\ell\}: \sum_\ell b_\ell = Lb} \sum_{\ell=1}^L 2^{-2b_\ell} v_\ell(\pi)^{1/g}, \quad (39)$$

and restore C_g in the optimal distortion at the end.

Optimal allocation and distortion. For fixed π , each term $2^{-2b_\ell} v_\ell^{1/g}$ is convex in b_ℓ and the constraint is affine, so first-order conditions are sufficient. Stationarity of the Lagrangian $\mathcal{L} = \sum_\ell 2^{-2b_\ell} v_\ell^{1/g} + \mu(\sum_\ell b_\ell - Lb)$ gives $2 \ln 2 \cdot 2^{-2b_\ell} v_\ell^{1/g} = \mu$ for all ℓ : the optimal allocation equalizes per-group distortions. Solving and enforcing the constraint,

$$b_\ell^* = b + \frac{1}{2g} \log_2 \left(v_\ell(\pi) / \prod_m v_m^{1/L}(\pi) \right), \quad (40)$$

which is (8). Substituting back, every term equals $2^{-2b} \left(\prod_m v_m(\pi) \right)^{1/(Lg)}$, so the inner minimum is

$$L 2^{-2b} \left(\prod_{\ell} v_{\ell}(\pi) \right)^{1/(Lg)} = L 2^{-2b} \left(\prod_{i=1}^d \sigma_i^2 \right)^{1/d}, \quad (41)$$

where the last equality uses $\prod_{\ell} v_{\ell}(\pi) = \prod_{i=1}^d \sigma_i^2$ for every partition and $Lg = d$. The value is independent of π ; restoring the common factor, the optimal distortion is $D^* = C_g L 2^{-2b} \left(\prod_{i=1}^d \sigma_i^2 \right)^{1/d}$, proving the theorem.

Fixed rates. Under $b_{\ell} \equiv b$, the distortion is $C_g 2^{-2b} \sum_{\ell} v_{\ell}(\pi)^{1/g}$. By the arithmetic–geometric mean inequality applied to the nonnegative numbers $v_{\ell}^{1/g}$,

$$\sum_{\ell=1}^L v_{\ell}^{1/g} \geq L \left(\prod_{\ell} v_{\ell}^{1/g} \right)^{1/L} = L \left(\prod_{i=1}^d \sigma_i^2 \right)^{1/d}, \quad (42)$$

with equality iff all v_{ℓ} coincide, so the fixed-rate distortion is at least $C_g L 2^{-2b} \left(\prod_i \sigma_i^2 \right)^{1/d} = D^*$. The equal-rate distortion attains the unconstrained optimum on volume-equalizing partitions. ■

G.2 Low-Rate Version of the Grouping Bound

The remark in Sec. 4 removes the high-resolution assumption for independent Gaussian groups. Let $\mathbf{r}_{G_{\ell}} \in \mathbb{R}^{1 \times g}$ have independent Gaussian entries with variances $\{\sigma_i^2\}_{i \in G_{\ell}}$, and let $\mathcal{Q}_{\ell}$ be *any* quantizer with gb_{ℓ} bits. The rate of a fixed-rate quantizer upper-bounds the mutual information between source and reconstruction, so the distortion is bounded below by the distortion-rate function of the source. For an independent Gaussian vector, reverse water-filling (Berger, 1971) gives, when the entry-level distortion D/g is below $\min_{i \in G_{\ell}} \sigma_i^2$,

$$R(D) = \sum_{i \in G_{\ell}} 1/2 \log_2 \frac{g \sigma_i^2}{D} = \frac{g}{2} \log_2 \frac{g v_{\ell}^{1/g}}{D}. \quad (43)$$

Setting $R(D) \leq gb_{\ell}$ and solving for D ,

$$\mathbb{E} \|\mathbf{r}_{G_{\ell}} - \mathcal{Q}_{\ell}(\mathbf{r}_{G_{\ell}})\|_2^2 \geq g v_{\ell}^{1/g} 2^{-2b_{\ell}}. \quad (44)$$

The dependence on the partition is only through $v_{\ell}^{1/g}$, so the fixed-rate argument of Sec. G.1 applies to this lower bound.

H High-Resolution Model: Empirical Test

High-resolution quantization theory is often used as a design principle for compressors deployed at finite rates (Sullivan and Wiegand, 1998). The accuracy of the approximation depends on whether the quantization cells are small relative to variations in the source density and distortion measure, which can hold even at modest nominal rates. We emphasize, in any case, that our results are asymptotic rather than finite-rate guarantees. We test, at and around the deployed rate, the specific predictions that determine our design.

We focus on Theorem 2, which rests on the high-resolution distortion model (7),

$$D_{\ell}(b) = C_g 2^{-2b} v_{\ell}(\pi)^{1/g}, \quad (45)$$

while the method operates at $b = 2$, where the asymptotics are not guaranteed to hold. For a group G_{ℓ} of g transform entries, a partition π , and a rate b , we define the empirical distortion as

$$\widehat{D}_{\ell}(b) = \mathbb{E}_{\text{test}} \left[\|\mathbf{r}_{G_{\ell}} - \mathcal{Q}_{\ell}(\mathbf{r}_{G_{\ell}})\|_2^2 \right]. \quad (46)$$

We estimate this quantity using a 2^{gb} -entry k -means codebook trained on training-split coefficients and evaluated on held-out coefficients. We vary b unless otherwise stated and set $g = 4$ (matching the deployed

method). Here $\mathbb{E}_{\text{test}}$ denotes the empirical mean over held-out coefficients, and $\widehat{D}_{\ell, \text{train}}$ denotes the same estimate on the training split.

We evaluate (45) using parameters estimated from the training data. The group volume is

$$v_{\ell}(\pi) = \prod_{i \in G_{\ell}} \sigma_i^2, \quad (47)$$

where the coordinate variances σ_i^2 are estimated on the training split. Let $b_0 = 2.5$. We estimate the common constant once as

$$\widehat{C}_g = 2^{2b_0} \text{median}_{\ell} \left(\widehat{D}_{\ell, \text{train}}(b_0) / v_{\ell}(\pi)^{1/g} \right), \quad (48)$$

where the median is taken over the groups of the equalizing partition and all tested (layer, KV head) pairs. We hold $\widehat{C}_g$ fixed across all held-out rates, groups, and partitions. In the empirical comparison, we therefore evaluate

$$D_{\ell}(b) = \widehat{C}_g 2^{-2b} v_{\ell}(\pi)^{1/g}. \quad (49)$$

We also report the volume-normalized empirical distortion $\widehat{D}_{\ell}(b) / v_{\ell}(\pi)^{1/g}$, which the model predicts to be approximately common across groups at a given rate:

$$D_{\ell}(b) / v_{\ell}(\pi)^{1/g} = \widehat{C}_g 2^{-2b}. \quad (50)$$

Here the hat denotes an empirical estimate and is unrelated to the reconstruction notation of Appendix A.

We test three predictions. First, at each rate, the volume-normalized distortions $\widehat{D}_{\ell}(b) / v_{\ell}(\pi)^{1/g}$ should approximately agree across groups. Second, across rates, the volume-normalized distortion should follow the scaling 2^{-2b} . Third, at the deployed rate $b = 2$, the measured total distortion $\sum_{\ell=1}^L \widehat{D}_{\ell}(b)$ and the theoretical partition criterion $\sum_{\ell=1}^L v_{\ell}(\pi)^{1/g}$ should order partitions in the same way.

H.1 Setup

We used Qwen3-8B, Qwen3-4B-Thinking-2507, and Llama-3.1-8B post-RoPE keys at three (layer, KV head) pairs across the model depth (layer 1, head 0; layer 17, head 3; layer 35, head 7 for the Qwen3 models, and layer 1, head 0; layer 15, head 3; layer 31, head 7 for the 32-layer Llama-3.1-8B), giving $3 \times L = 96$ groups per partition per rate at $d = 128$, $g = 4$. Coefficients are $\mathbf{r} = (\mathbf{k} - \bar{\mathbf{k}})\mathbf{R}_K$ with the deployed transform. Per-token RMS normalization was off for this study; it isolates the unnormalized transform coefficients analyzed in Theorem 2. Throughout, b denotes the rate for the key branch in bits per entry. For Qwen3 we pooled three model captures (GPQA-198, MMLU, GPQA-32), each 8 sequences of 16384 tokens; examples 0-5 of each capture formed the training pool, from which we retained $2^{18} - 1 = 262143$ coefficient vectors, and examples 6-7 formed the held-out set (98304 tokens); the training and held-out sets contain disjoint sequences from each capture. For Llama-3.1-8B, we used a single LongBench capture of 80 variable-length prompts split 0-23 / 24-47 (262144 training, 98304 held-out vectors). Codebooks were initialized by sampling 2^{g^b} codewords uniformly without replacement from the training vectors, followed by 25 Lloyd iterations; empty cells were reseeded to the training points worst served by the current codebook. We consider rates $b \in \{1.0, 1.5, 2.0, 2.5\}$, i.e. $2^{g^b} \in \{16, 64, 256, 1024\}$ codewords. In testing the partitions, we consider the three methods in Appendix A: equalizing (deployed), random, and variance-sorted.

H.2 Agreement across groups

At each rate, dividing each group’s distortion by $v_{\ell}^{1/g}$ removes most of the variation across the 96 groups (Fig. 8a–b). The volume term accounts for the bulk of the per-group differences, which is the component of the model that the partition argument of Theorem 2 relies on.

H.3 Rate dependence

We fit the generalized rate law $2^{-\alpha b}$ to the volume-normalized held-out distortions at $b \in \{1, 1.5, 2, 2.5\}$ and obtain $\alpha = 1.816$ for Qwen3-8B, $\alpha = 1.828$ for Qwen3-4B-Thinking-2507, and $\alpha = 1.878$ for Llama-3.1-8B,

compared with the high-resolution value $\alpha = 2$. If we compute the exponent between adjacent rates, which is more relevant for an asymptotic law, we obtain 1.796, 1.813, and 1.840 on held-out data for the intervals $1 \rightarrow 1.5$, $1.5 \rightarrow 2$, and $2 \rightarrow 2.5$ on Qwen3-8B, increasing monotonically over the tested range, a trend consistent with approaching the asymptotic value 2. The fitted constants agree closely across the three models, $\hat{C}_g = 1.518$, 1.507, and 1.489 respectively, as expected for a quantity that should depend only on g and the source shape. As we observe in Fig. 8a, 8d, and 8g, with $\hat{C}_g$ estimated from training data at the top rate ($b = 2.5$), the measured and predicted curves are closest at that rate and separate toward lower rates.

Because these distortions are measured on held-out coefficients, the reported deviation includes both finite-rate effects and any residual suboptimality of the trained vector quantizers; we do not attempt to separate these effects.

H.4 Partition comparison

At $b = 2$, Table 13 shows the measured and predicted per-group distortion over the 96 groups for each partition. The predicted criterion ranks the three partitions in the same order as the held-out distortion, with the deployed equalizing partition performing best. The model is quantitatively accurate for the equalizing and random partitions, but overestimates the distortion of the variance-sorted partition by 81.5%.

Table 13 Partition comparison at $b = 2$ for Qwen3-8B. Measured and predicted mean per-group distortion over 96 groups. Predictions use (7) with $\hat{C}_g = 1.518$.

Partition	Measured	Predicted	Pred./meas.
Equalizing	0.739	0.803	1.087
Random	0.941	1.060	1.125
Variance-sorted	2.610	4.740	1.815

Residual dependence. We observe that our model provides worse predictions for the variance-sorted partition. Beyond existing finite-rate approximation errors, we argue that this effect is consistent with a stronger violation of the independent-Gaussian assumption. The transform decorrelates the entry at second order but does not make them independent; the sorted partition places adjacent entries together, and those groups exhibit substantially greater dependence as measured by Shannon total correlation $TC = \sum_j h_j - h_{\text{joint}}$ (Table 14, Kozachenko–Leonenko k -NN joint-entropy estimates and m -spacing marginal estimates on training coefficients). We remark that total correlation indicates a violation of the independence assumption but does not quantify its effect on distortion. Finite-sample entropy estimates may yield slightly negative estimated total correlations, explaining why a reported mean can be below the corresponding median.

Table 14 Total correlation per 4-coordinate group for Qwen3-8B on training coefficients.

Grouping	Mean	Median	q90
Equalizing	0.0290	0.0647	0.1077
Random	0.0419	0.0643	0.1488
Sorted	0.2284	0.0906	0.4488

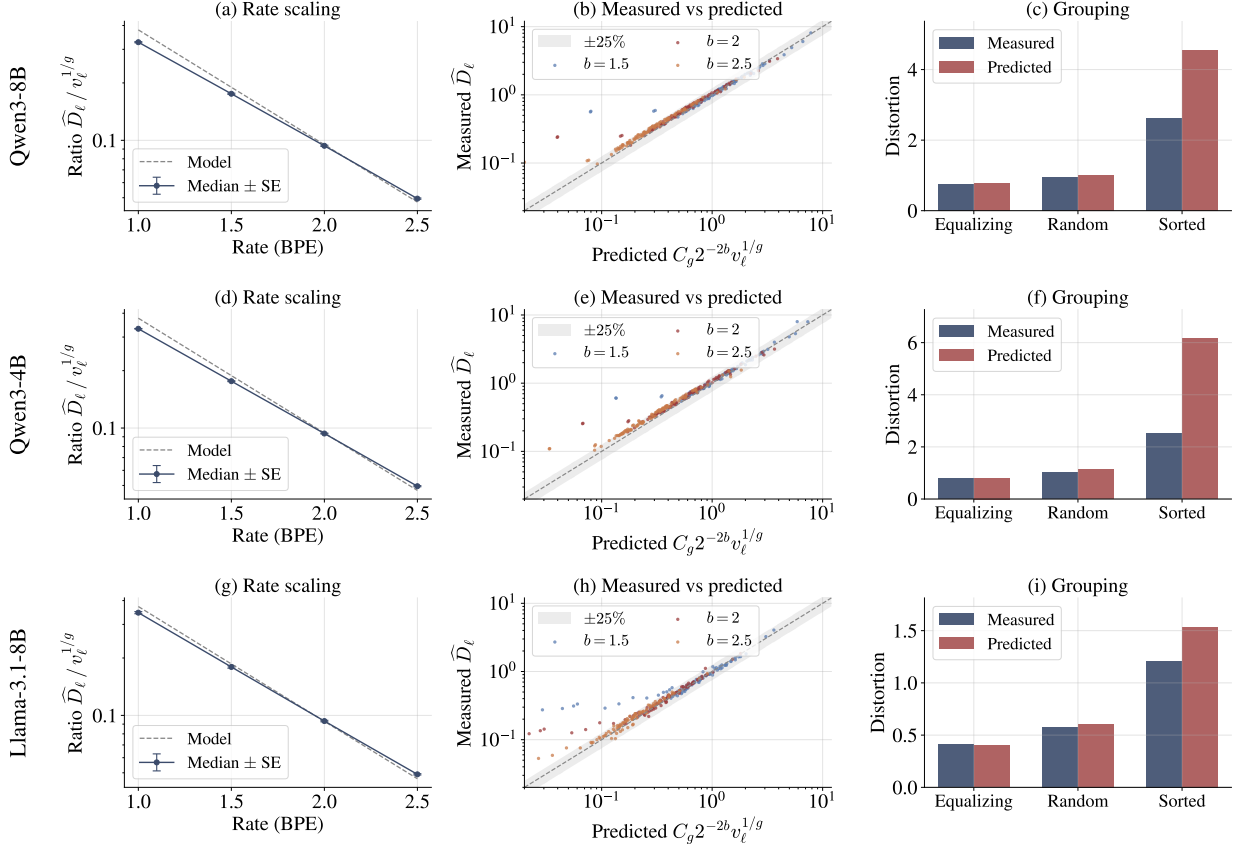


Figure 8 High-resolution model test, 96 groups pooled over three (layer, KV head) pairs of Qwen3-8B (a-c), Qwen3-4B-Thinking-2507 (d-f), and Llama-3.1-8B (g-i). (a, d) Median volume-normalized distortion $\widehat{D}_\ell / v_\ell^{1/g}$ against rate, with the model curve $\widehat{C}_g 2^{-2b}$ tied to the median at $b = 2.5$; the curves are closest at the anchor and separate at low rate, where the local exponent is below 2. (b, e) Per-group measured distortion $\widehat{D}_\ell$ against the model D_ℓ at $b \in \{1.5, 2, 2.5\}$; the band is $\pm 25\%$ around the identity. (c, f) Measured and predicted per-group distortion for the three partitions at $b = 2$; the ordering is preserved.

I Further evaluations

RULER NIAH measures exact-match retrieval. To test whether the comparison transfers beyond that setting, we evaluate on two suites: LongBench (Sec. I.1), a multitask suite of QA, summarization, and code completion, and LooGLE (Sec. I.2), long-document QA and summarization with longer inputs and a larger item count.

I.1 LongBench

We evaluate on LongBench (Bai et al., 2023) tasks (Qasper, QMSum, MultiNews, TREC, TriviaQA, SAMSum, LCC, RepoBench-P) using the full test sets (200 examples per task; 500 for LCC and RepoBench-P) and greedy decoding (default for this benchmark), so runs are deterministic. Prompt KV entries are quantized during prefill at 2.28 BPE for OSCAR and 2.22 BPE for NOVA-KV, while KV entries generated during decoding remain in BF16; layer 0 is kept in full precision for every arm. OSCAR and NOVA-KV differ only on the key quantizer: both share the same SQ value path. NOVA-KV transforms and codebooks are calibrated on the same GPQA-Diamond subset we used in the main paper.

Qwen3-8B We let both OSCAR and NOVA-KV use the sink-plus-recent band ($n_{\text{sink}}=64$, $n_{\text{rec}}=256$). We also ablate the effect of this band: the half-band rows reduce it to $n_{\text{sink}}=32$, $n_{\text{rec}}=128$ for both methods. Since decoding is deterministic, we assess separability with a paired bootstrap over items (10000 resamples,

percentile 95% CIs); CIs on the mean row are computed by resampling items within each task and averaging the eight per-task means with equal weight, so each task counts 1/8 regardless of test-set size. Per-task intervals are uncorrected for multiple comparisons and are provided as descriptive analyses; the equal-weight average is the primary comparison.

Table 15 LongBench scores per task for Qwen3-8B. Full band is $(n_{\text{sink}}, n_{\text{rec}}) = (64, 256)$; half band is $(32, 128)$. BF16 is band-independent. Best compressed method per task and band setting in **bold**.

Task	BF16	Full band		Half band	
		OSCAR	NOVA-KV	OSCAR	NOVA-KV
Qasper	44.27	43.72	39.65	41.63	39.96
QMSum	24.67	23.28	23.72	22.87	23.58
MultiNews	24.92	24.40	23.70	24.29	23.34
TREC	41.50	49.50	46.00	36.50	50.00
TriviaQA	90.53	87.85	89.74	88.35	89.74
SAMSum	40.00	39.75	37.73	39.78	37.66
LCC	64.90	60.79	63.52	60.83	62.85
RepoBench-P	60.05	50.93	61.75	49.87	61.38
Mean	48.86	47.53	48.23	45.52	48.56

Table 16 Paired bootstrap comparison, gap between NOVA-KV and OSCAR, per task and averaged. Entries are Δ with 95% CI; * marks CIs excluding zero.

Task	Full band	Half band
Qasper	−4.07 [−7.63, −0.61]*	−1.67 [−5.73, 2.30]
QMSum	+0.44 [−0.31, 1.17]	+0.71 [0.04, 1.40]*
MultiNews	−0.70 [−1.12, −0.28]*	−0.95 [−1.39, −0.50]*
TREC	−3.50 [−10.50, 3.50]	+13.50 [7.00, 20.50]*
TriviaQA	+1.90 [−0.77, 4.83]	+1.40 [−1.90, 4.75]
SAMSum	−2.02 [−3.36, −0.70]*	−2.12 [−3.57, −0.71]*
LCC	+2.73 [1.05, 4.41]*	+2.02 [0.25, 3.80]*
RepoBench-P	+10.82 [8.51, 13.19]*	+11.51 [9.13, 13.86]*
Average	+0.70 [−0.40, 1.79]	+3.05 [1.90, 4.20]*

Table 17 Paired bootstrap comparison against the BF16 reference, using equal-weight average over the eight tasks. * marks CIs excluding zero.

Arm	Average Δ	95% CI
OSCAR, full band	−1.33	[−2.33, −0.29]*
NOVA-KV, full band	−0.63	[−1.47, 0.26]
OSCAR, half band	−3.34	[−4.40, −2.24]*
NOVA-KV, half band	−0.29	[−1.26, 0.65]

LongBench contexts are shorter than the regime where the methods separate on RULER NIAH, so the differences in average score are small. First, against BF16 (Table 17), we detect no significant difference between NOVA-KV and the uncompressed reference at both band settings, while OSCAR is significantly below it at both. Second, at the full band, NOVA-KV’s +0.70 advantage over OSCAR in average score is not separable from zero (Table 16). Third, at the half band the average-score gap becomes separable (+3.05): halving the BF16 band costs OSCAR 2.0 points on average but leaves NOVA-KV almost unchanged, indicating that NOVA-KV is less dependent on protecting sink and recent tokens.

GPT-OSS-20B. To check whether the OSCAR failure on GPT-OSS (cf. Sec. J.2) is specific to RULER NIAH’s exact-match retrieval, we separately evaluate the same eight LongBench tasks on GPT-OSS-20B using the

same configuration we used for Qwen3-8B (in this case, we use the full protection band, $n_{\text{sink}} = 64, n_{\text{rec}} = 256$). Table 18 reports every task. OSCAR collapses on *every* task (question answering, summarization, classification, and code completion alike) for a mean of 7.51 against BF16’s 36.76, an 80% relative drop. NOVA-KV stays within 7% of BF16 (34.24), consistent with the Qwen3-8B result in Table 15.

Table 18 LongBench (Bai et al., 2023) scores per task for GPT-OSS-20B. Best compressed method per task in **bold**.

Task	BF16	OSCAR	NOVA-KV
Qasper	36.92	4.84	31.58
QMSum	19.62	6.37	19.32
MultiNews	20.64	10.04	18.89
TREC	17.00	2.50	11.00
TriviaQA	76.58	2.67	72.07
SAMSum	28.18	3.22	25.75
LCC	50.82	17.57	51.69
RepoBench-P	44.29	12.84	43.63
Mean	36.76	7.51	34.24

I.2 LooGLE

We evaluate on LooGLE (Li et al., 2023) with Qwen3-8B (1101 items), scored by ROUGE-L, under the same serving configuration and band as the LongBench evaluation (Sec. I.1); decoding is greedy, so runs are deterministic. Separability is assessed with the same paired bootstrap over items (10000 resamples, percentile 95% CIs).

Table 19 LooGLE ROUGE-L for Qwen3-8B over $n=1101$ items, with paired bootstrap differences (95% CIs; * marks CIs excluding zero).

Arm	ROUGE-L
BF16	31.50
OSCAR	26.38
NOVA-KV	28.81
<i>Paired differences</i>	
NOVA-KV – OSCAR	+2.43 [1.07, 3.78]*
NOVA-KV – BF16	−2.69 [−3.79, −1.62]*
OSCAR – BF16	−5.12 [−6.60, −3.69]*

Discussion. NOVA-KV scores 2.43 ROUGE-L above OSCAR, separable from zero, and 2.69 below the BF16 reference, roughly half of OSCAR’s 5.12-point degradation. Unlike on LongBench, where NOVA-KV was statistically indistinguishable from BF16, the larger item count in this dataset resolves all pairs. LooGLE complements RULER NIAH by measuring graded long-document QA and summarization rather than exact-match retrieval.

J GPT-OSS-20B: Quantizing a Hybrid-Attention Cache

As attention computation and KV-cache size become the dominant cost of long-context generation, recent architectures increasingly adopt hybrid designs in which only a fraction of the layers attend globally over the full context, and the remaining layers use a cheaper mechanism. The mechanism varies across models: sliding-window attention (Team et al., 2025; Agarwal et al., 2025), gated delta networks (Yang et al., 2025b), or state-space layers (Lieber et al., 2024). The consequence for KV quantization is the same in each case: only the global layers’ cache grows with context, so the cache that a quantizer sees is smaller, the information it holds is denser, and there is less redundancy, limiting potential compression gains. GPT-OSS-20B (Agarwal et al., 2025) is such a model and differs from Llama-3.1-8B and Qwen3-8B in every property relevant to KV

quantization (Table 20): only 12 of its 24 layers attend globally (the other 12 see a 128-token window), heads are half as wide, keys are not bounded by QK-norm, every head carries a learned sink logit that competes with tokens in the softmax, and the feed-forward blocks route each token to 4 of 32 experts. Its growing per-token cache is 5–6 $\times$ smaller than the dense models’. We quantize only the 12 global-attention layers; the sliding-window layers stay BF16, since their cache is bounded by the window and does not grow with context. In this setting, NOVA-KV is the only 2-bit method (among the methods we tested) that remains effective (Table 2). This section describes the serving changes the architecture requires, analyzes why the scalar baselines fail, and measures the rate at which scalar quantization recovers.

Table 20 Architecture properties relevant to KV quantization. “KV bytes/token” counts the BF16 cache that grows with context (global-attention layers).

	Llama-3.1	Qwen3-8B	GPT-OSS
layers (global / total)	32 / 32	36 / 36	12 / 24
sliding window	–	–	128 (12 layers)
head dim.	128	128	64
heads (Q / KV)	32 / 8	32 / 8	64 / 8
QK-norm	no	yes	no
attention sink	no	no	learned
feed-forward	dense	dense	MoE (4 of 32)
KV bytes/token	131072	147456	24576

J.1 Attention sinks in the quantized serving path

The learned sink enters the softmax as an extra logit per head that contributes to the normalizer but has no value vector. Three changes are required in the quantized read path.

Split-KV normalization. In the two-stage decode kernel (Sec. B.5), each KV split reduces independently. The sink term $\exp(s_h - m)$ is added to the normalizer exactly once, after the cross-split combine in stage 2, not per split.

Mean centering and offset invariance. NOVA-KV stores centered keys and drops the constant $\mathbf{q}^\top \bar{\mathbf{k}}$ from the logits, which is harmless under a softmax (Lemma S1, Sec. F.1). With a sink the invariance fails: token logits carry the offset but the learned sink logit does not, so the sink’s softmax share would be rescaled by $\exp(\mathbf{q}^\top \bar{\mathbf{k}})$. The kernel therefore shifts the sink logit by the same per-query constant (one d -dimensional dot product per KV head per step), which restores exact equivalence.

Chunked prefill. New chunks are scored against the dequantized prefix by a flash-attention kernel (Dao et al., 2022) with no sink support; its output and log-sum-exp are rescaled afterwards to fold the sink into the normalizer.

J.2 On the OSCAR and QuaRot failures

On GPT-OSS-20B, OSCAR collapses on retrieval (Table 2) and QuaRot collapses on every task, while NOVA-KV stays within a few points of BF16 at the same rate. We test whether this is an integration artifact or a property of the model.

Serving stack. We run the model under plain HuggingFace and overwrite the cache with the same codec the pool applies. On a 10-needle greedy retrieval probe at 1K context, GPT-OSS drops from 4 needles (its BF16 control on this harness) to 1 with keys and values quantized, and to 0 with only keys quantized; Llama-3.1-8B retrieves 10 of 10 both in BF16 and with OSCAR under the identical procedure. Since the value-side OSCAR codec is shared with NOVA-KV, which is unaffected, the failure is specific to SQ of the keys.

Reconstruction fidelity. We measure the OSCAR codec on real captured keys and queries in four bases (Table 21). GPT-OSS keys *reconstruct better* than Llama’s in the served bases (9.1 vs. 8.2 dB in the Hadamard basis, 8.8 vs. 8.1 dB in the calibrated one), and the rows are not outlier-dominated there: the mean excess kurtosis of a key row is -0.3 on GPT-OSS and -0.1 on Llama in the calibrated basis (versus 20.8 and 7.2 in the identity basis, where both models are heavy-tailed and both SNRs collapse). The same key error nevertheless produces $3.5\text{--}3.9\times$ the relative attention-logit error on GPT-OSS, and the ratio persists across all four bases. The architecture concentrates all long-range information in 12 narrow-head layers, so each cached element carries more task information and the logits tolerate less noise: one scale per (token, head) at 2 bits sits above GPT-OSS’s noise tolerance and below Llama’s. The MoE feed-forward adds an amplification path: the perturbed attention output enters the router, whose top- k selection is a discrete function of the hidden state. In a teacher-forced probe (24 decode steps at 1K context, production codec), KV quantization changed the top-4 expert set at 70% of (token, layer) router decisions, an error mode a dense feed-forward does not exhibit.

Table 21 OSCAR key quantization on GPT-OSS-20B (G) and Llama-3.1-8B (L): reconstruction SNR of the keys and relative attention-logit error $\|\mathbf{q}^\top(\mathbf{K} - \hat{\mathbf{K}})\|/\|\mathbf{q}^\top\mathbf{K}\|$, per basis. Post-RoPE keys and queries captured from GPQA prompts (4 prompts for the SNR statistics, 8 for the logit error), averaged over all quantized layers and KV heads; per-(token, head) scale with the production percentile clip (0.96). “Calibrated” is each model’s served rotation.

Basis	Key SNR (dB)		Logit error	
	G	L	G	L
calibrated	8.8	8.1	0.64	0.17
Hadamard	9.1	8.2	0.59	0.16
random	—	—	0.69	0.17
identity	3.0	5.8	0.89	0.44

QuaRot. The QuaRot rows in Table 2 already include the protective BF16 sink-plus-recent band and still sit at 0.0. Without the band, i.e. unclipped min-max QuaRot-INT2 on every token from position 0, GPT-OSS produces degenerate output at every prompt length we probed, from 90 to 8000 tokens (greedy completions; served and reproduced in the HuggingFace simulation), while Llama only degrades under the identical procedure. On this model the unclipped scalar codec has no usable operating point, with or without the band.

J.3 Scalar quantization recovers at higher rates

If information density is the correct explanation, a modest amount of additional rate should restore the scalar codec. We simulate the OSCAR codec bit-exactly on the write path of a BF16 server (values are dequantized before storage; no memory is saved, and every token is quantized) and vary the rate (Table 22). One additional bit takes the codec from collapse to functional, and 4 bits matches BF16 on retrieval. Finer scale granularity at 2 bits does not substitute, despite a higher rate than INT3: the deficit is resolution per coordinate, not outlier range. NOVA-KV reaches the required fidelity at 2.22 BPE through vector quantization (NIAH-8K 89.6, Table 2), consistent with the main-text result that, among the techniques we tested, it is the only 2-bit method that remains effective on this model.

Table 22 Scalar (OSCAR-codec) rate ladder on GPT-OSS-20B, simulated bit-exactly in serving. “Bits” is the payload width of every stored key and value coordinate; “groups” is the number of min–max scale/zero-point pairs per (token, head) row of $d=64$ coordinates (1 is the served configuration). Single greedy rollout; NIAH-8K on the 200-prompt balanced subset (25 per subtask), MATH-500 on 100 prompts. BPE charges BF16 scale and zero-point per group ($b + 0.5g$).

Bits	Groups	BPE	NIAH-8K	MATH-500
2	1	2.5	0.8	4
2	4	4.0	27.0	15
3	1	3.5	82.8	73
4	1	4.5	98.3	85
BF16		16	99.0	92

K A Chunked-Prefill Dequantization Leak

Long prompts are prefilled in chunks rather than in one pass, bounding peak prefill memory and letting the scheduler interleave a long prompt’s prefill with other requests’ decoding. In a mixed-precision KV pool, each chunk’s keys and values are quantized as they are written, so a later chunk of the same prompt attends over earlier chunks’ dequantized rows instead of their original values – a leak that single-pass prefill does not have.

The leak can be closed by keeping an exact copy of a request’s own in-flight rows until its final chunk completes, after which cached content matches the quantized-on-write baseline exactly. This is not free: while a request’s own prefill is in progress, its in-flight rows occupy close to BF16-sized cache rather than the compressed footprint, trading part of the memory saving for correctness during that window.

Table 23 reports RULER NIAH with the leak present and closed. It is real and grows with context (Llama OSCAR: +4.5 at 8K, +16.6 at 128K), and closing it recovers most of the gap for NOVA-KV and for Llama’s OSCAR. GPT-OSS’s OSCAR does not recover: 2–11 across all five lengths versus a floor of 0 with the leak present. The leak is real but does not explain the collapse (Sec. J.2).

Table 23 RULER NIAH accuracy with the chunked-prefill leak present → closed. One seed, 25 prompts per NIAH subtask (200 total). OSCAR and NOVA-KV on GPT-OSS-20B, and, as a control, Llama-3.1-8B.

Length	GPT-OSS-20B		Llama-3.1-8B	
	OSCAR	NOVA-KV	OSCAR	NOVA-KV
8K	0.50 → 11.38	89.75 → 92.75	82.25 → 86.75	92.88 → 95.00
16K	0.00 → 4.50	79.62 → 92.75	78.62 → 86.12	91.75 → 94.50
32K	0.00 → 2.38	80.00 → 90.62	80.50 → 86.38	88.62 → 93.25
64K	0.00 → 4.25	69.25 → 85.00	74.38 → 85.12	87.25 → 93.38
128K	0.00 → 2.00	54.50 → 67.88	31.12 → 47.75	54.25 → 65.75

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